Al-Muneer Online Portal

Event Venue Management Systems - Web

AHMED HANI GHALEB

UNIVERSITI TEKNOLOGI MALAYSIA

UNIVERSITI TEKNOLOGI MALAYSIA
DECLARATION OF PROJECT REPORT

Author's full name	:	Ahmed Hani Ahmed Ghaleb			
Student's Matric No.	:	A21EC9120	Academic Session	:	24/25 2
Date of Birth	:	2001/04/19	UTM Email	:	ahmedahmed@graduate.utm.my
Project Report Title	:	AL-MUNEER ONLINE PORTAL			

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DEDICATION

This thesis is dedicated to my father, who taught me that the best kind of knowledge to have is that which is learned for its own sake. It is also dedicated to my mother, who taught me that even the largest task can be accomplished if it is done one step at a time.

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ABSTRACT

Al-Muneer Hall for Weddings and Events, situated in Ibb, Yemen, currently handles bookings and inquiries mainly through manual phone calls and direct messages. This conventional method results in a considerable time commitment from the owner in responding to repetitive inquiries and managing processes by hand, which often overwhelms the staff and may lead to delays or inaccuracies for clients. Additionally, prospective customers, especially female stakeholders who adhere to local cultural norms and may prefer remote assessments, do not have access to a centralized, visually engaging platform to thoroughly explore the venue and manage their interactions effectively. The Al-Muneer Online Portal tackles this issue by offering a dedicated, all-encompassing web-based platform that consolidates all vital information and key interaction processes into an easily accessible, user-friendly interface. Built using contemporary web development techniques, the portal will include an interactive availability calendar, comprehensive pricing details, an extensive FAQ section, and a high-quality visual gallery. Importantly, it will also feature capabilities for users to submit booking inquiries, optionally upload proof of payment (such as transfer screenshots) to validate bookings, and share feedback regarding their experience. For the administrator, the system will provide a secure panel to oversee all content, manage availability and pricing, review submitted payment proofs, handle feedback, and manage notifications by generating pre-filled WhatsApp messages to facilitate direct communication with clients. This strategy is designed to create a centralized, dependable source of information and interaction, available around the clock, thus significantly alleviating the burden of manual inquiries and processes, enhancing the hall's marketing visibility, streamlining operations, and improving the overall customer experience with cultural awareness.

ABSTRAK

Dewan Al-Muneer untuk Perkahwinan dan Acara, yang terletak di Ibb, Yaman, pada masa ini menguruskan tempahan dan pertanyaan terutamanya melalui panggilan telefon manual dan mesej langsung. Pendekatan tradisional ini menyebabkan pemilik perlu meluangkan masa yang signifikan untuk menjawab soalan berulang dan menguruskan proses secara manual, yang sering membebankan staf dan berpotensi menyebabkan kelewatan atau ketidaktepatan maklumat kepada pelanggan. Tambahan pula, bakal pelanggan, terutamanya pihak berkepentingan wanita yang mengikut norma budaya tempatan dan mungkin lebih suka membuat penilaian dari jauh, kekurangan platform berpusat yang kaya dengan visual untuk meneroka dewan secara menyeluruh dan menguruskan interaksi mereka dengan cekap. Portal Dalam Talian Al-Muneer menangani cabaran ini dengan menyediakan platform berasaskan web yang khusus dan komprehensif, menyatukan semua maklumat penting dan proses interaksi utama ke dalam antara muka yang mudah diakses dan mesra pengguna. Dengan memanfaatkan amalan pembangunan web moden, portal ini akan menampilkan kalendar ketersediaan interaktif, maklumat harga terperinci, Soalan Lazim (FAQ) yang komprehensif, serta galeri visual berkualiti tinggi. Yang penting, ia juga akan menggabungkan fungsi untuk pengguna menghantar pertanyaan tempahan, secara pilihan memuat naik bukti pembayaran (cth., tangkapan skrin pindahan) untuk mengesahkan tempahan, dan memberikan maklum balas mengenai pengalaman mereka. Bagi pentadbir, sistem ini akan menawarkan panel selamat untuk mengurus semua kandungan, ketersediaan, harga, melihat bukti pembayaran yang dihantar, mengurus maklum balas, dan mengurus pemberitahuan dengan menjana mesej WhatsApp pra-isi bagi memudahkan komunikasi langsung dengan pelanggan. Pendekatan ini bertujuan untuk menyediakan sumber maklumat dan interaksi berpusat yang boleh dipercayai, boleh diakses 24/7, sekali gus mengurangkan beban pertanyaan dan proses manual secara signifikan, meningkatkan kehadiran pemasaran dewan, melancarkan operasi, dan menambah baik pengalaman pelanggan secara keseluruhan dengan kepekaan budaya.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The Al-Muneer Hall for Weddings and Events (قاعة المنير للأفراح والمناسبات) is a place in Ibb, Yemen that provides services for weddings and social events in a particular community. As far as now, the operations in the hall are carried out based on traditional means of communication such as phone calling and messaging. Though it is a working system, it involves a lot of inefficiency on behalf of the owner and the potential customers as well, who may need some additional information or want to complete their booking procedure smoothly. Therefore, the suggested solution to the problem is the development of "Al-Muneer Online Portal" that is going to help the business owner to make his client interaction and operational processes easier and more efficient. The portal should be created as a user-friendly and aesthetically pleasing information system providing such functions as inquiry form, payments proofs uploading, feedback collection, reporting from the owner's side, and notifications for clients.

1.2 Problem Background

The current operation system of Al-Muneer Hall faces various challenges due to its reliance on manual communication. The owner spends too much time responding to recurring queries about the availability of the venue, prices, facilities, and pictures. Apart from handling such queries at the beginning, managing the booking process, which includes confirmation by verbal agreement and even recording of offline payments, collecting feedback systematically from customers, and obtaining business performance insight is done manually and is very tedious. All these tasks that require manual effort take up valuable time. Furthermore, the reliance on direct communication may lead to inaccuracies, limited access for the clients, and no systematic method to collect feedback to improve services.

A key part of this problem has its foundation in the socio-cultural setting of Ibb, Yemen. Possible clients, particularly women in general, do not have an easy way to check the place out and interact there. There is no centralized system for collecting information, making bookings through easy procedures such as uploading payment transfer screen shots (which are common practice in a mostly cash-based economy with limited bank transactions), and collecting feedback in an organized way. This approach is inefficient, unprofessional, and cannot make use of digital technology to market and manage the service effectively.

1.3 Project Aim

The core aim of this study will be the design and development of the Al-Muneer Online Portal, which is an internet based system meant to serve as the main interface and tool used by the Al-Muneer Hall for Weddings and Events. The aim of this will be to centralize information regarding the venues, streamline the process of inquiries and booking, create feedback and basic reporting capability for the owner of the business, notify the customers in time, relieve the owner from repetitive work, and create a better user experience.

1.4 Project Objectives

To achieve the project aim, the following objectives have been defined:

1. To elicit and analyze the system requirements for the Al-Muneer Hall Online Portal, including hall booking, availability checking, inquiries, payment proof submission, feedback, reporting, and notification management.
2. To design the system architecture, database, and user interface of the portal based on the analyzed requirements.
3. To develop the Al-Muneer Hall Online Portal in accordance with the produced design.
4. To test and validate the developed portal against the specified functional and non-functional requirements.

1.5 Project Scope

Al-Muneer Online Portal is a browser-based responsive application that has no mobile application and can be accessed on desktops and mobile phones. This application caters to two types of users; the first type is the visitors who can access the following features; viewing the venue details, browsing the media gallery, checking availability through calendar, reviewing pricing and frequently asked questions, making booking inquiries, uploading the proof of payments and giving feedback.

The other type is the admin (owner) who logs in the portal and manages the following; hall information, media gallery, availability of the hall, pricing information, FAQs, booking inquiries, verifying the proof of payments, managing feedbacks and basic operational reports. The notification feature of the application uses pre-filled WhatsApp messages.

The system covers the following functions; showing of the content, booking inquiries, tracking the availability, updating the price information, handling feedbacks and reporting. However, there are certain limitations in the system, which include lack of automatic upload of proof of payments, lack of connection to external calendars, live chat, CRM, advanced event management features and sophisticated analytics such as graphs. The initial content will be provided by the owner, and the traffic to be moderate, while the reports will be basic including the inquiry summary, bookings based on payment status and feedback management.

1.6 Project Importance

The creation of the Al-Muneer Online Portal holds great importance. First, it helps resolve operational problems in terms of automating the process of information dissemination and improving such processes like booking confirmations and getting feedback from clients. Moreover, it improves the user experience by providing 24/7 access to comprehensive information and advice concerning inquiries and bookings. What is more important, considering the cultural aspect of the country, this portal acts as an essential tool in making sure of inclusiveness and effective communication. Remote assessment becomes possible, and booking confirmation process is simplified using the payment proof methods familiar to locals. The introduction of the feedback mechanism makes it possible for the owner to improve the service quality continuously basing on client feedback. Simple reporting provides the owner with valuable insights into the operation process without any additional tools needed. Instant notification improves communication and increases the level of satisfaction of the clients.

1.7 Report Organization

The report consists of six unique chapters. Chapter 1 provides an introduction to the project, where the background of the problem, project objectives, aims, scope, and significance is given. Chapter 2 describes a literature review related to the sphere of online booking platforms and event venues management, covering the aspects of payment system, feedback, reports, notifications, user interface design principles and the technologies to be implemented. Chapter 3 gives more information about the methodology applied in the whole process of system development. Chapter 4 covers the aspects of system analysis and design, which include requirement specification, architecture, database design, and UI/UX design. Chapter 5 describes the results of the system implementation, the most important interfaces and testing procedures performed in PSM2. Finally, Chapter 6 summarizes the work done within the project, discusses the results and suggests possible improvements.

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CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

In this chapter, an overview of the latest literature and technology used in relation to the development of the Al-Muneer Online Portal is provided. This chapter explores the area of online venue management systems, emphasizing the change from the traditional way of managing events to modern systems offering greater capabilities. The review covers the key characteristics that are expected to be found in the modern venue portals, including the inquiry management systems, easy process of booking confirmation (providing the evidence of the payment made), feedback collecting system, simple reporting dashboard and notification system. In addition to that, this chapter provides a case study of Al-Muneer Hall in order to place the development of the system into context. The chapter analyzes the drawbacks of the existing process and compares the proposed system to the currently available solutions or alternative processes. In addition to that, this chapter covers the literature on the technology selected for the portal's development.

2.2 Project Domain

The Al-Muneer Online Portal is classified as a specific online portal that offers information, handling queries, and operation services in relation to the sectors of event management and hospitality. The purpose of such systems is to bridge the gap of communication between the venues providers and the potential customers in an efficient manner compared to traditional systems (Sigala et al., 2012; Buhalis & Law, 2008).

2.2.1 Key Features and Information Architecture for Event Venue Portals

The modern-day event venue websites are no longer merely static brochures, but have rather taken up an interactive form. The literature discusses several key features that make a website successful, such as good quality images, transparency of prices, and calls-to-actions for lead generation (Nielsen, 2000). Availability calendars that would provide instant information should be one of the key aspects. In those environments where it is rare to find online payment facilities, it would be useful to provide an opportunity for a user to prove payment made offline (Kshetri, 2013).

2.2.2 Manual Operation Analysis

Current manual system at Al Muneer Hall is common among SMEs but suffers from the followings:

1) Time Consumption: Staff spend time on repetitive queries and manual tracking of bookings, payments, and feedback (Laudon & Laudon, 2016).
2) Limited Availability: Information and process progression are tied to staff hours.
3) Inconsistency: Information can vary; tracking booking confirmations and feedback lacks standardization.
4) Challenges in Scalability: Issues with managing growing volume.
5) Poor User Experience: Customers want to have simple online experiences such as clear steps to confirm and ways to give feedback (Sigala et al., 2012)
6) Documentation and Oversight Challenges: Manual tracking makes it hard to generate operational insights or reliably follow up on client interactions.

2.2.3 Current System Analysis - Comparison of Existing Systems/Approaches

To make a comparative analysis, it is important to describe the categories of the current systems or methodologies that are available for a venue like Al-Muneer Hall to implement. The current manual method of interaction of Al-Muneer Hall involves a lot of direct communication between the staff of the organization and the customers. The communication part is done largely through phone calls, personal meetings and other messaging tools like WhatsApp. This approach allows for a lot of customization but it is very labor intensive and has some limitations on scaling, consistency of information distribution and maintaining the records (Harrigan et al., 2012). Booking, payments, and feedback collection are informal and systemless.

The other type is general online booking sites such as Eventbrite or Cvent. Many opportunities exist with such third-party platforms for hosting event listing, ticketing and sometimes even full event management. They offer users a handy interface to search and book events and services offered. These platforms have a lot of options but they may have their own branding, fees and workflow that do not necessarily align with the interests of an independent venue (Gazzaroli et al., 2019). They are more on ticketed events and not customized venue rental requests.

There are also user-friendly website builders like Wix, Squarespace or GoDaddy Website Builder that help to create a website relatively easily by using predefined templates and drag-and-drop technology (Odoom et al., 2017). These sites are good for simple presence and presentation of the company on the web. However, the built-in abilities to perform dynamic actions like creating a custom calendar of availability, generating customized prices for complex services, collecting payment proof, or any kind of reporting are often limited unless enhanced by third-party plug-ins, which further adds complexities and costs.

And finally there are also quite popular standard static websites of competitors in local businesses. Basically, these websites are online brochures that provide visitors some basic contact information, a gallery and sometimes a list of services. Usually there are no interactive elements to check availability, get dynamic pricing, send structured queries or do any transaction related work online.

A comparative analysis helps to position the proposed Al-Muneer Online Portal:

Table 2.1: Comparative Analysis of Existing Systems and the Proposed Portal

Feature	**Al-Muneer Online Portal (Proposed)**	**Manual (Phone/WhatsApp)**	**Generic Booking Platforms**	**Simple Website Builders**	**Static Competitor Website**
Venue-Specific Info	Tailored	(Manual)	Generic Structure	(Manual Setup)	Basic
Interactive Calendar	Yes	No	Often (Complex Setup)	Limited/ Plugin Dependent	Often No
Dynamic Pricing Display	Yes (Admin Managed)	(Verbal/ Manual)	Sometimes (Complex)	Usually No	Often No
Admin Panel	Custom Built	N/A	Yes (Platform Specific)	(General CMS)	N/A
Payment Proof Upload	Yes (Simple)	(Manual Exchange)	Often Not Direct	No	No
Feedback System	Yes	(Ad-hoc)	Sometimes (Platform)	Usually No	No
Basic Reporting	Yes	No	Limited (Platform)	No	No
Notifications	Manual (prefilled messages)	(Manual)	Email Often Primary	No	No

Cultural Context Adapt	High (Custom Design)	High (Direct Interaction)	Low (Standardized)	Medium (Template Dependent)	Medium

2.2.4 Case Study: Al-Muneer Hall for Weddings and Events

In order to contextualize the project in the real-world environment, the following section provides some information on Al-Muneer Hall, the organization where the system will be developed. Al-Muneer Hall (قاعة المنير للأفراح والمناسبات) is an event hall located in Ibb, Yemen. The main activity of this company is the provision of venues for wedding parties, which will help satisfy the needs of the local community and maintain the cultural traditions of Yemeni people associated with these celebrations. Moreover, this hall can provide venues for other kinds of social events such as funerals and celebration parties. Al-Muneer Hall is a single-location business. As far as its structure is concerned, it should be mentioned that Al-Muneer Hall is a family-owned business. The primary stakeholder of the business is Mr. Ahmed Almunajid who is the owner of the company, is involved in the company activities, and communicates with clients. While there might be more staff members helping in organizing the event, cleaning, and other aspects of event hosting, it should be noted that such functions as administration and reservation and communication with customers are performed by Mr. Almunajid or his manager.

In terms of engaging with the stakeholder for the purpose of gathering requirements, the main technique was engaging directly with the owner, Mr. Ahmed Almunajid.

First off, there was a discussion arranged through video conference (Google Meet) on April 4, 2025, in order to get to know the business and find out what kind of issues they have and what they want in the online portal. In addition to that, there was a continuous correspondence via WhatsApp for additional details (evidences are provided in communication logs and NABC proposal document (Appendix B)). Based on the findings obtained from such engagement which can be regarded as an interview with the user, it turns out that a major amount of time of Mr. Almunajid is consumed with answering repetitive phone calls related to availability, prices, hall capacity and details. Moreover, there was a need for a professional web site in order to show the hall. In addition, there were specific demands concerning the functionality: there should be an interactive calendar, the way to show prices clearly, media gallery and FAQ section. At present, the process of booking is done via calls or personal visits and usually payments are made in cash but in some cases there can be local bank transfers with screenshot proof of payment. At the moment, there is no system for managing those proofs or any feedbacks at all. Preferences in communication are mostly WhatsApp and calls rather than emails. One of the key needs was the system which could be easily managed and used both by the owner and the clients. After the discussion of the initial proposal and feedback, Mr. Almunajid was ready to introduce a way to upload payment proofs by the clients, a possibility to leave a comment, basic reporting tools and prefilled notifications, ideally WhatsApp.

2.3 Literature Review of Technology Used

The selection of technologies for the Al-Muneer Online Portal is based on their suitability for developing a robust, scalable, and maintainable web application that meets the defined requirements, including the newly added functionalities. For each core technology, alternatives were considered.

Table 2.2: Technology Stack Selection and Justification

Technology	Purpose	Alternatives Considered	Justification
Spring Boot (Java) / MVC	Backend framework	Python (Django/Flask), Node.js (Express.js), PHP (Laravel)	Spring Boot offers a mature, convention-over-configuration implementation of the MVC pattern, embedded server deployment, and strongly typed compile-time checking through Java, which reduces runtime defects in form-driven transactional applications (Walls, 2016). Its integrated ecosystem (Spring Data JPA, Spring Security) covers persistence and authentication requirements without third-party dependencies, and the MVC separation of concerns improves maintainability and testability (Fowler, 2002).
PostgreSQL	Relational database	NoSQL (MongoDB, Cassandra), SQLite, MS SQL Server	SQLite, MySQL, MS SQL Server|The system's data — bookings, availability slots, pricing tiers, payment proofs — is highly structured, interrelated, and requires transactional integrity (e.g., a booking must atomically reserve its calendar slot). A relational DBMS providing full ACID guarantees and referential-integrity enforcement is therefore the appropriate model, whereas NoSQL stores target schema-flexible or horizontally partitioned workloads not present here (Garcia-Molina et al., 2008). PostgreSQL was selected over the alternatives for its standards-compliant SQL, proven reliability under concurrent writes, and open-source licence with no acquisition cost, which suits the stakeholder's budget constraints. SQLite was rejected for its

			limited concurrency, and MS SQL Server for its licensing cost.
HTML5, CSS3, JS (Responsive)	Frontend development	JavaScript Frameworks (React, Angular, Vue.js)	The portal is a content-centric, server-rendered application with limited client-side state; a single-page-application framework would introduce a build toolchain, larger payloads, and framework upkeep without corresponding benefit. Server-side rendering with standards-based HTML5/CSS3 yields faster first-page loads on the low-bandwidth mobile connections typical of the target user base, better search-engine indexability, and accessibility conformance (W3C, 2018; Nielsen, 2000). Minimising the technology surface also reduces long-term maintenance effort for a system that will be maintained by a single administrator.
Waterfall Model	SDLC methodology	Agile (Scrum, Kanban), Spiral Model	The requirements were fully elicited and signed off with the stakeholder before design began, the scope is fixed, and the project follows the sequential PSM1/PSM2 academic timeline with document-gated milestones (SRS, SDD, STD). Under these conditions of stable requirements and a fixed schedule, the sequential Waterfall model provides clear phase deliverables and traceability, whereas iterative methods are advantageous chiefly when requirements are volatile (Royce, 1970)
Cloud Hosting (VPS)	Deployment environment	AWS, serverless	A fixed-specification VPS provides a predictable flat monthly cost, which the stakeholder's budget requires, in contrast to usage-metered cloud services whose costs vary with traffic (Armbrust et al.,

			2010). The portal's anticipated moderate traffic fits comfortably within a single VPS instance, and full root access simplifies deploying a self-contained Spring Boot artifact alongside its PostgreSQL database. Managed cloud and serverless offerings add elasticity the workload does not need, at the cost of pricing unpredictability and vendor lock-in (Kshetri, 2013).
Admin Auth (Hashed Pass.)	Secure admin login	Plain text storage (highly insecure), Reversible encryption	Passwords are stored as salted one-way BCrypt hashes, the approach recommended by industry password-storage guidance, so that credentials cannot be recovered even if the database is compromised (Schneier, 1996). Plain-text storage offers no protection, and reversible encryption concentrates risk in the encryption key; a deliberately slow adaptive hash additionally resists offline brute-force attacks.
Notification Mechanisms	(e.g., WhatsApp Deep-linking, Email libraries, Manual WhatsApp Messaging)	Notification Mechanisms (WhatsApp Deep-linking, pre-filled messages)	dmin-to-client notifications|In-app notifications (via web dashboard), Full WhatsApp Business API|WhatsApp is the dominant communication channel among the target clientele, so notifications delivered there have far higher reach than an in-app dashboard that clients would need to revisit. Pre-filled deep links deliver templated messages through the admin's existing WhatsApp account at zero integration and subscription cost, whereas the WhatsApp Business API requires provider onboarding, per-conversation fees, and hosting of

			webhook infrastructure — recurring costs disproportionate to a single-venue business's notification volume. The deep-link approach satisfies the stakeholder's requirement of admin-initiated, personally reviewed messages while remaining upgradeable to the full API if volume grows

The Model-View-Controller (MVC) architecture, enabled by Spring Boot, guarantees a clear separation of concerns, thereby simplifying the management and scalability of the application (Fowler, 2002). Relational database systems such as PostgreSQL ensure data integrity, which is essential for the effective management of bookings and venue details (Garcia-Molina et al., 2008). Implementing responsive design through HTML5, CSS3, and JavaScript is imperative for contemporary web applications to accommodate a variety of user devices (W3C, 2018). Although Agile methodologies are widely embraced, the structured approach of the Waterfall model can be beneficial for projects with well-defined initial requirements, as described in the proposal phase (Royce, 1970). Ensuring security through HTTPS and appropriate password hashing is critical for maintaining user trust and safeguarding data (Rescorla, 2001; Schneier, 1996). Cloud hosting provides both flexibility and reliability for application deployment (Armbrust et al., 2010).

2.4 Chapter Summary

The chapter has presented a comprehensive literature review on the Al-Muneer Online Portal. This chapter has identified the scope of the project in terms of online event venue management system and discussed important features that have been mentioned in previous researches, including visual-rich design, availability calendar, price listing, feedback form, easy booking process, basic reports generation capabilities, and notification system. Limitations of the traditional manual process have been analyzed in light of the existing situation at Al-Muneer Hall and the necessity of the transition to the digital world has been emphasized. Also, a case study based on Al-Muneer Hall has been conducted in order to describe stakeholders involved in the project and corresponding business requirements. Various types of existing systems, such as booking systems and website building systems, have been analyzed and compared with the custom portal being developed. Finally, this chapter has reviewed the chosen technology stack and discussed other possible options.

CHAPTER 3

SYSTEM DEVELOPMENT METHODOLOGY

3.1 Introduction

The development of the Al-Muneer Online Portal having requirements and scope specified needs to follow a systematic and structured methodological approach so as to ensure that all objectives are accomplished in an efficient and effective manner. Right from the first concept and then through the planning and design phases associated with this PSM1 stage, the methodology that will be followed by the project should be such that it will take it through all of the entire life cycle before taking it further towards development and deployment in the PSM2 stage. System development methodology to be followed for the development of the Al-Muneer Online Portal, its various phases and methodology implementation for the accomplishment of objectives of the project have been discussed in this chapter. A suitable methodology for a project of this nature where requirements are already known and structured approach is needed is Waterfall methodology.

3.2 Methodology Choice and Justification

Given the linearity and sequential process of software development using the Waterfall model, this model is chosen for the development of the Al-Muneer Online Portal project. This model requires that there be segmentation of the project life cycle into phases and completion of one phase prior to moving on to the next. Given that there has been some preliminary coordination and clarification regarding the proposal, the Al-Muneer Online Portal project is well positioned to benefit from such a rigid model since it is suitable for projects whose requirements have been specified from the onset of the project. The distinct scope of functionalities that include an interactive calendar, prices displayed, uploading payment proofs, feedback, ease of reporting and notification make this model appropriate.

Life cycles in the waterfall approach should typically be used in cases where predictability and levels of documentation are key criteria. The clear identification of the stages in the Waterfall approach ensures established indicators to measure progress and outcomes such as SDD and SRS in terms of FYP student project with defined academic milestones (Progress 1, Progress 2, Final Report). Agile methods allow flexible approaches to project execution especially when there is a change in requirements; however, the Al-Muneer Online Portal is better off with the Waterfall approach in light of its strong emphasis on early planning and designing, thus ensuring lower uncertainty while shifting from the planning and designing phase in PSM1 to developing phase in PSM2. Such an approach will ensure the certainty that the developing process in PSM2 can proceed upon a solid base that has been agreed on by all the parties involved in the project.

3.2.1 Waterfall Model Process Diagram

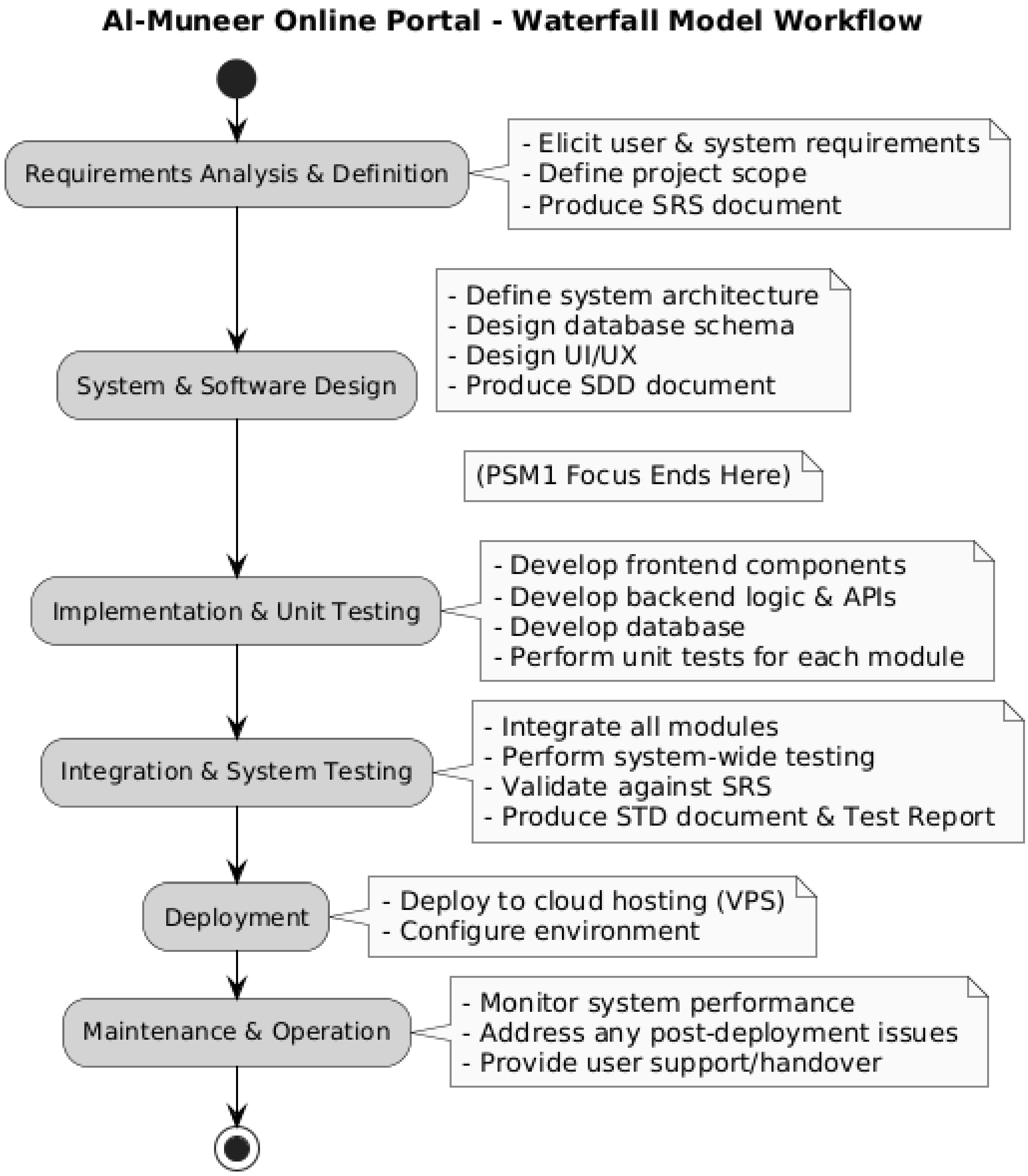


Figure 3.1: Waterfall Model Workflow for the Al-Muneer Online Portal

3.3 Phases of the Chosen Methodology

The development life cycle of Al-Muneer Online Portal based on the Waterfall model has the different phases. Each phase has its own goals and products to be delivered and they lead in an orderly way to the final product.

1. **Requirements Analysis and Definition:** This stage is very important to understand and capture the special needs of the Al-Muneer Online Portal. Detailed discussions with stakeholder (Mr. Almunajid) are included for capturing all functional and non-functional requirements. The involved activities are determination of users' roles (visitor and admin), listing of functionalities which include presentation of venue details, media gallery, availability calendar, ability to customize pricing, submission of booking requests, uploading of payments, providing feedback, administration of content by the admin, provision of payment status updates, review of feedbacks, generation of basic reports and setting up of notifications. The primary output of this phase is the Software Requirement Specification (SRS). This stage was one of the main activities in the PSM1 stage.

2. **System and Software Design:** Once requirements are defined and documented in SRS document, design phase begins. In this phase total system architecture of Al-Muneer Online Portal is designed. It includes designing the database schema (MySQL/PostgreSQL), designing the backend (Spring Boot services/APIs) and designing the UI and UX for the front end for visitors and the back end for the administration. Decisions about module segregation, flow of data, and security measures are determined during this phase. The main output of this phase is the System Design Document (SDD), which acts as a guideline for the development phase (student developer). This is an essential part of the PSM1 process, especially before Progress 2.

3. **Implementation and Unit Testing:** This is the point where real coding and development commence, and most of it will be done in PSM2. As per the

SDD, the Al-Muneer Online Portal will be developed. The web application will be built in one integrated project. The backend logic will be handled using Spring Boot (Java), and will also act as the UI for the portal. The View layer will be designed using server-side templating engine (Thymeleaf) together with HTML, CSS and JavaScript. The database will be set up and populated with some structures. Each component will be tested as it is being developed in order to confirm its functionality.

4. **Integration and System Testing:** Once the units are completed and tested individually, they are integrated to form a complete system. The complete system then goes through system testing to ensure that all components work well together as a system and that the entire system meets the requirements stated in the SRS. This includes functional testing, usability testing, basic performance testing, and security testing. An STD will be used for this purpose, and a test report will be compiled from it. This stage is also meant to take place in PSM2.

3.3.1 Gantt Chart for PSM1

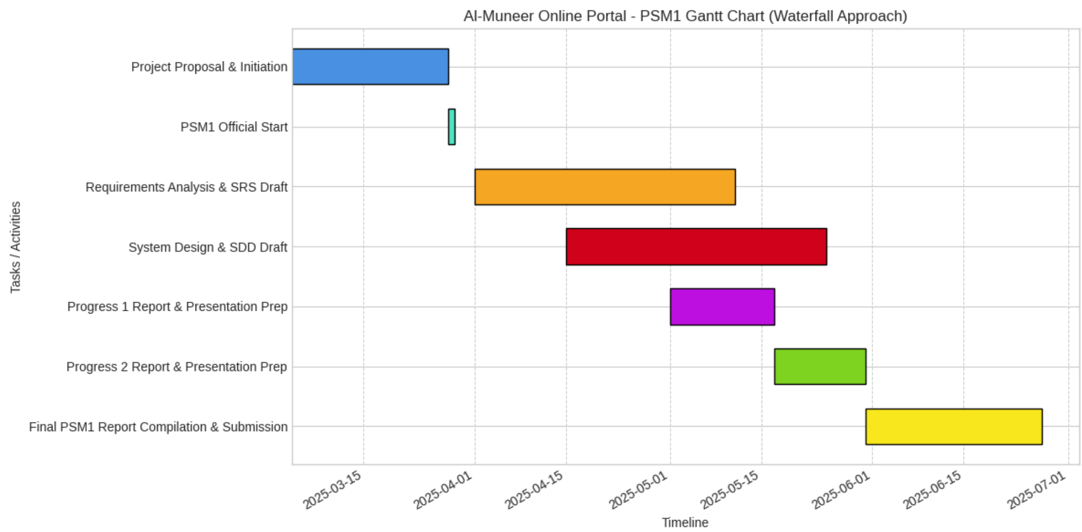


Figure 3.2: Gantt Chart for PSM1 Activities

3.3.2 Gantt Chart for PSM2

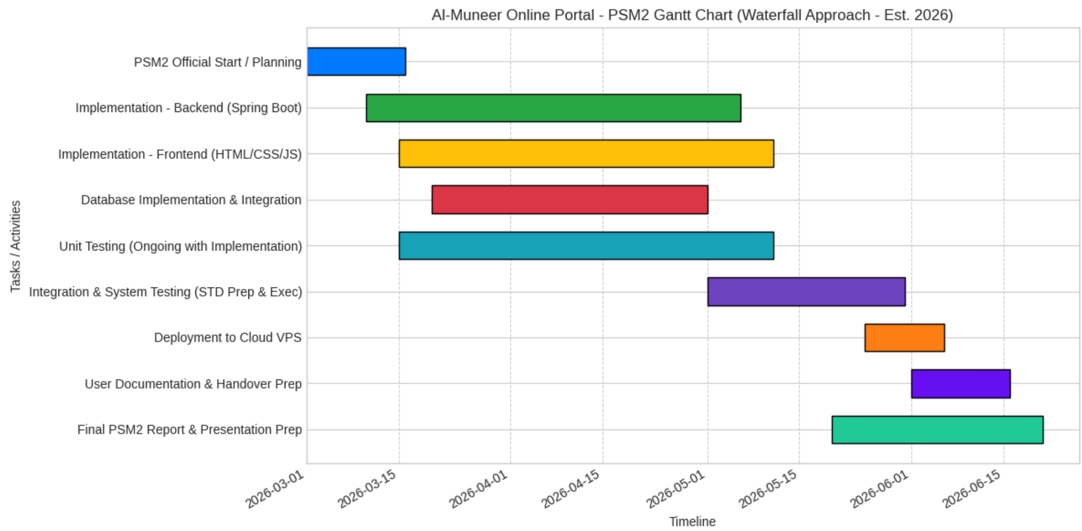


Figure 3.3: Proposed Gantt Chart for PSM2 Activities

3.4 System Requirements Analysis: Hardware and Software

The successful development and deployment of the Al-Muneer Online Portal will require certain hardware and software resources. The main hardware requirement for the development process will be a relatively new PC or laptop. The machine should be equipped with the following features: decent processor (Intel Core i5 or AMD Ryzen 5, or better), at least 8GB of RAM (preferably 16GB to work comfortably with development tools), as well as enough hard drive space (at least 256GB SSD). Software requirements for the development process include a suitable operating system (Windows, Mac OS, or Linux), a Java Development Kit (JDK) version compatible with Spring Boot (e.g., JDK 11, 17, or 21), a development environment such as IntelliJ IDEA, Eclipse, or VS Code with Java plugins, a build tool such as Maven or Gradle (provided by Spring Initializr), a relational database management system (locally installed MySQL Community Server or PostgreSQL), a web browser for testing (Chrome or Firefox), and a version control software (Git).

Hardware: In the case of the deployment environment, a Cloud Virtual Private Server (VPS) would be used. The VPS should have specifications, which include at least 1-2 vCPUs, 2GB-4GB RAM, and enough SSD storage (e.g., 20GB-50GB) in order to support the application, database, as well as all files uploaded (e.g., payment proofs). The operating system of the server should normally be some kind of Linux OS (e.g., Ubuntu or CentOS). In terms of software for the deployment server, it would be required to have the Java Runtime Environment (JRE) compatible with the application, chosen relational database server (MySQL or PostgreSQL), web server or application server, if the embedded server of Spring Boot is not used (though Spring Boot uses embedded servers like Tomcat, Jetty, Undertow). Also, the set of useful tools for server administration, monitoring and security (e.g., firewall, SSH) would be needed. Moreover, an SSL/TLS certificate would be required in order to use HTTPS.

3.5 Technology Used Description

Al-Muneer Online Portal's development with the use of the Waterfall model entails systematic inclusion of a number of technologies in each phase of the development process. During the phase of System & Software Design (PSM1), the architecture describes the interactions between those technologies. For instance, the backend built on Spring Boot (Java) based on MVC architecture is to host the core business logic. That includes the functionality for managing venues details, booking requests, confirmation of payments, feedback, reporting, and notifications.

At this stage the relational database MySQL/PostgreSQL will be created to hold all of the permanent data, such as user passwords (hashed), hall description, availability, pricing policies, inquiries, payment proofs and feedback entries. This structured system will allow the implementation of the data fetching process necessary for the administrator reporting capabilities, directly.

The user interface will be created as the View part of the MVC architecture. It will use HTML5, CSS3 and JavaScript and will render server-side with a templating engine (Thymeleaf for example) controlled by Spring Boot. This methodology makes an extremely tight application where the backend can directly control the UI, which is exactly what this project aims to do.

The view will be constructed within the Model-View-Controller design pattern as the User Interface. The View will employ HTML5, CSS3, and JavaScript languages, and will involve server-side rendering using a template engine like Thymeleaf controlled by the Spring Boot. Such an approach creates an application that is tightly coupled where the backend has control over the User Interface, something that matches the needs of this project very well.

During the implementation stage, there will be a use of Cloud VPS where the packaged application of Spring Boot together with the database will be hosted. HTTPS (SSL/TLS) communication protocol will be required to ensure that all data exchanged between client browsers and the server remains secure. Waterfall model ensures that each of these technological elements undergoes design, implementation, and testing process systematically and in order.

3.6 Chapter Summary

The purpose of this chapter is to give a complete description of the System Development Methodology that has been used in the Al-Muneer Online Portal. This chapter has emphasized the importance of the Waterfall model in the implementation of the system development methodology. The reasons why this methodology has been adopted in this project is due to the fact that it follows a sequential approach and this makes it suitable where there are clearly defined requirements and deliverables. All the various phases of this methodology have been discussed in details including Requirements Analysis and Definition up to Deployment and

Maintenance, assigning particular tasks to PSM1 and PSM2. Visual Gantt Charts have been utilized for defining the timeframe of each phase. This chapter also described the hardware and software requirements of the development and deployment environment of the portal. It has been shown how the technologies that have been chosen such as Spring Boot, relational database, and web frontend technology will work in the development process according to the waterfall methodology.

CHAPTER 4

REQUIREMENT ANALYSIS AND DESIGN

4.1 Introduction

The requirements and design of the Al-Muneer Online Portal are described in this chapter, along with the description of the system architecture, database structure, principles of user interface design, and required processes. The primary goal of this stage, conducted within PSM1, is the transformation of the requirements of the stakeholders and goals of the project defined in previous stages into a thorough design plan of the system in PSM2. This design is oriented towards providing an intuitive and efficient system to the Visitor, who is interested in information about Al-Muneer Hall, and Owner, who is responsible for the operation of the portal.

4.2 Requirement Analysis

Requirement analysis consists of identifying the user needs as well as the attributes of the system. For the Al-Muneer Online Portal, the requirements were mainly obtained by engaging with the stakeholder, Mr. Almunajid, who is the owner of the Al-Muneer Hall, as discussed in the Case Study (Chapter 2.2.4). Furthermore, a study was performed on the usual features expected of a similar online event management portal for the Yemen environment.

4.2.1 Functional Requirements

Functional requirements describe what the system should do. For the Al-Muneer Online Portal, these are categorized by user roles and key system operations:

FR1: Visitor Functionalities

1) **FR1.1 View Venue Viewing:** The system will show complete venue information such as the description, capacity, offered services, and location information.
2) **FR1.2 View Media Gallery:** The system will let visitors to view a gallery of images and possibly videos of the venue.
3) **FR1.3 Check Availability:** The system will offer an interactive calendar where visitors can see which days have already been booked and which are free.
4) **FR1.4 View Pricing Information:** The system will show price information. It could be basic prices, as well as additional charges depending on what services visitors choose.
5) **FR1.5 View FAQ:** The system will display a list of Frequently Asked Questions along with their answers.
6) **FR1.6 Submit Booking Inquiry:** The system will offer a form for visitors to submit an inquiry, including their name, contact information, date(s) and other specific requests.
7) **FR1.7 Submit Payment Proof (Optional):** Submitting Payment Proof (optional): The system will give visitors an option, either before submitting an inquiry or at the admin's request, to submit a picture (a screenshot of a local transfer receipt, for example) as evidence of their payment. The system will accept only common image formats (for instance, JPG, PNG) and set a file size limit.
8) **FR1.8 Submit Feedback:** The system shall provide a form for visitors to submit feedback about the venue or their experience with the portal.

FR2: Administrator Functionalities

1) **FR2.1 Secure Admin Login:** The system shall require administrators to enter their login information with unique, authenticated credentials (username and hashed password). The system shall display an error message in case of authentication failure.
2) **FR2.2 Manage Venue Information:** Manage Venue Information The system shall allow for creation, updating and deleting of text information about the hall (descriptions, services, FAQs, etc.) by the administrator.
3) **FR2.3 Manage Media Gallery:** It will enable the administrator to add images and videos to the gallery and remove those already added.
4) **FR2.4 Manage Availability Calendar:** It will enable the admin to identify the dates as booked or free in the interactive calendar.
5) **FR2.5 Manage Pricing:** The administrator will have the ability to set and change base pricing and customizable pricing options.
6) **FR2.6 Manage Booking Inquiries:** This will allow the administrator to manage the booking inquiries that have been sent.
7) **FR2.7 Manage Payment Proofs:** It will allow the administrator to manage the payment proofs that are uploaded and change the booking status (like "Payment Verified" or "Pending Confirmation").
8) **FR2.8 Feedback Management** The system shall enable the administrator to see the feedback submitted by the user.
9) **FR2.9 Generate Basic Reports:** The administrator shall be able to view basic reports such as a list of inquiries in a date range, a summary of booking statuses, or an overview of feedback received.
10) **FR2.10 Manage Notifications:** The system will allow the administrator to manage the message templates pre-set for notifications such as inquiry received and payment proof verified, and to communicate using preferred modes, especially through WhatsApp triggers, while email will be a fallback mode.

11) **FR2.11 Generate AI Insights:** The system shall generate concise, data-grounded advisory insights on the administrator's reports and analytics pages by submitting the current operational metrics to an

external AI service (Google Gemini) and displaying the returned recommendations, with a graceful fallback message when the service is unavailable.

FR3: General System Functionalities

- **FR3.1 Responsive Design:** The system shall be responsive and accessible on common web browsers across various devices (desktops, tablets, smartphones).
- **FR3.2 Secure Data Handling:** Safe Data Processing: The system will process all data submitted by the users, which includes personal data from the inquiry and payment proof upload and feedback.
- **FR3.3 Notification Generation:** Based on admin configuration and specific system events (e.g, new inquiry, payment proof submission), the system shall generate notifications to be sent to relevant parties (client or admin).

4.2.2 Non-Functional Requirements

The non-functional requirements define the quality characteristics of the system.

- **NFR1 Performance:** The system shall load pages in an acceptable time frame (e.g., 3-5 seconds on a typical internet connection). The interactive calendar should be responsive to user
- **NFR2 Usability:** The site is easy to use and intuitive for the visitors and administrator. The preferred language will be Arabic with an alternative English language option in case it is feasible to implement under the current scope of the project.
- **NFR3 Security:**
    - **NFR3.1 Admin Authentication:** The admin login shall be protected against common security vulnerabilities. Passwords should be hashed.
    - **NFR3.2 Data Protection:** Users data provided by users and data uploaded by users (payment proofs) should be protected against unauthorized access.

- ○ **NFR3.3 SSL/TLS:** Client-to-server communication must be encrypted using HTTPS.

- **NFR4 Reliability:** The portal must be generally available for its time, without being unavailable.
- **NFR5 Maintainability:** Code should be well structured and commented and follow the chosen MVC pattern to allow for future updates or modifications (relevant for PSM2 and beyond).
- **NFR6 Scalability (Basic):** NFR6: Not anticipating having a very high number of concurrent users from the get go, the platform architecture (Spring Boot and selected database) need to be scalable to handle a decent amount of increase in visitors (10,000 visitors per month).

4.2.3 Use Case Table and Diagram

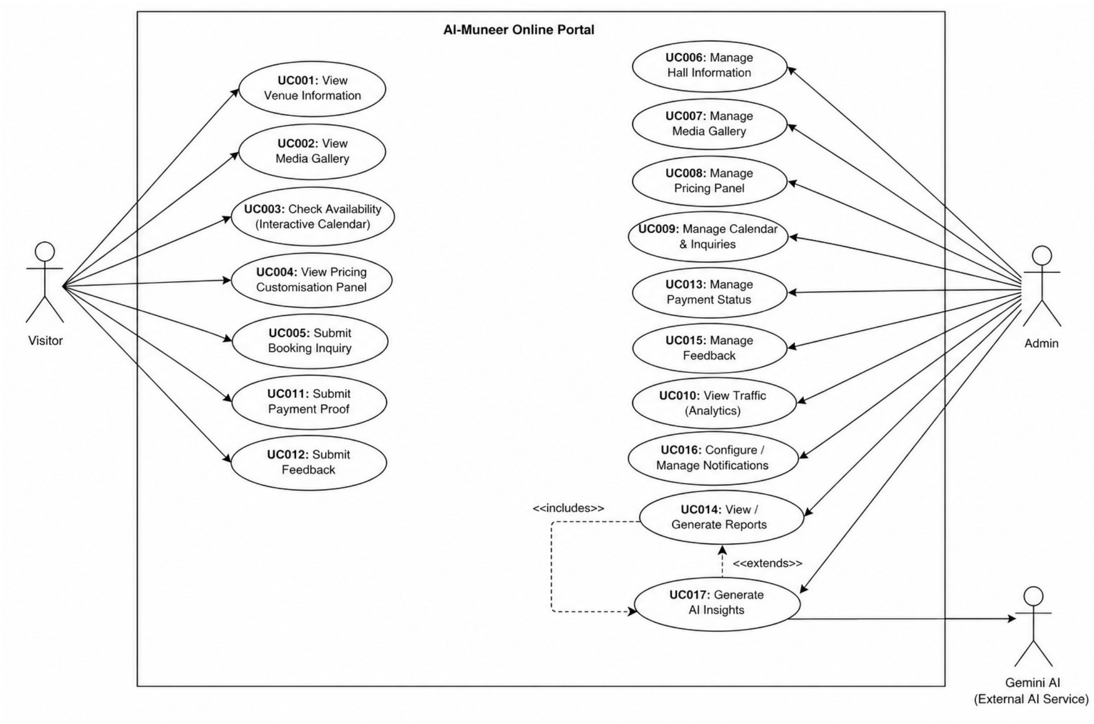


Figure 4.1: Use Case Diagram for the Al-Muneer Online Portal

4.2.3.1 Selected Use Case: Submit Booking Inquiry (UC005)

Table 4.1: Use Case Description for 'Submit Booking Inquiry' (UC005)

Attribute	Description
Use Case ID	UC005
Use Case Name	Submit Booking Inquiry
Actor(s)	Visitor
Pre-condition(s)	Visitor is on the Al-Muneer Online Portal and has navigated to the booking section/form.
Normal Flow	1. Visitor accesses the booking form. 2. Visitor fills in required fields (e.g., name, contact, desired date, event type, number of guests, specific requests). 3. Visitor clicks the "Submit Booking" button. 4. System validates the input data. **[If validation fails → AF1]** 5. System saves the booking details to the database. **[If the save operation fails → EF1]** 6. System displays a success message with the booking reference code to the Visitor. 7. System triggers a notification to the Administrator about the new booking. **[If notification delivery fails → AF2]**
Alternative Flow	1. AF1 — Invalid Input Data (triggered at Normal Flow step 4): 2. AF1.1. System displays an error message indicating the fields with invalid data.

	3. AF1.2. Visitor corrects the data 4. AF1.3. Flow resumes at Normal Flow step 3. 5. AF2 — Notification Failure (triggered at Normal Flow step 7): 6. AF2.1. System logs the notification failure; the booking remains saved and the success message remains displayed. 7. AF2.2. Administrator sees the new booking upon next login. Use case ends.
Exception Flow	1. EF1 — Database Save Error (triggered at Normal Flow step 5): 2. EF1.1. System displays an error message stating that the booking could not be submitted. 3. EF1.2. Visitor may retry submission (flow resumes at Normal Flow step 3) or abandon the use case. Use case ends.
Post-condition(s)	Booking is successfully recorded in the system. Administrator is notified of the new booking. Visitor receives confirmation with a reference code.

4.2.3.2 Sequence Diagram (Submit Booking Inquiry - UC005)

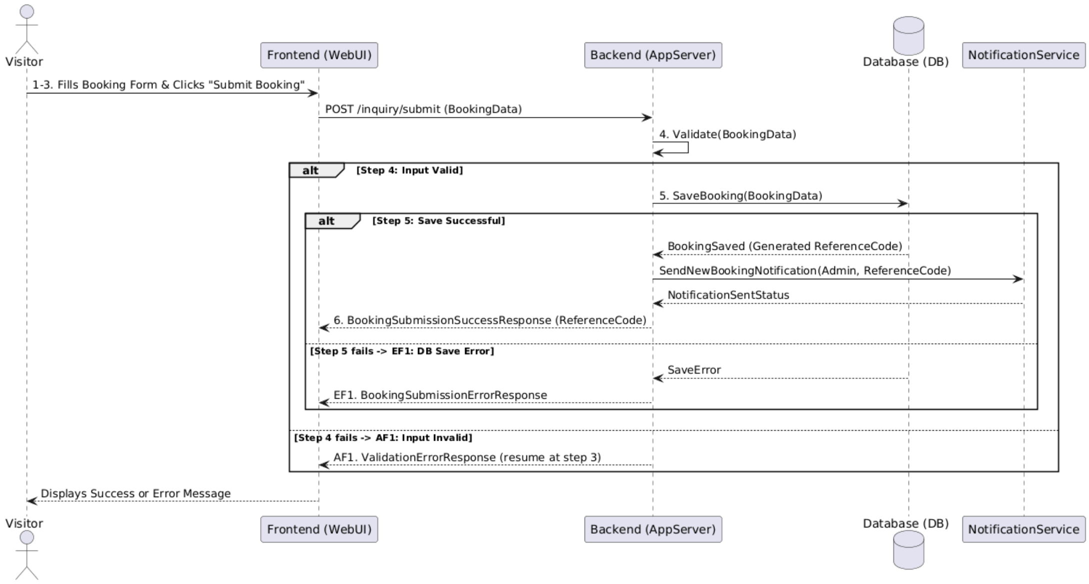


Figure 4.2: Sequence Diagram for 'Submit Booking Inquiry' (UC005)

4.2.3.3 Activity Diagram (Submit Booking Inquiry - UC005)

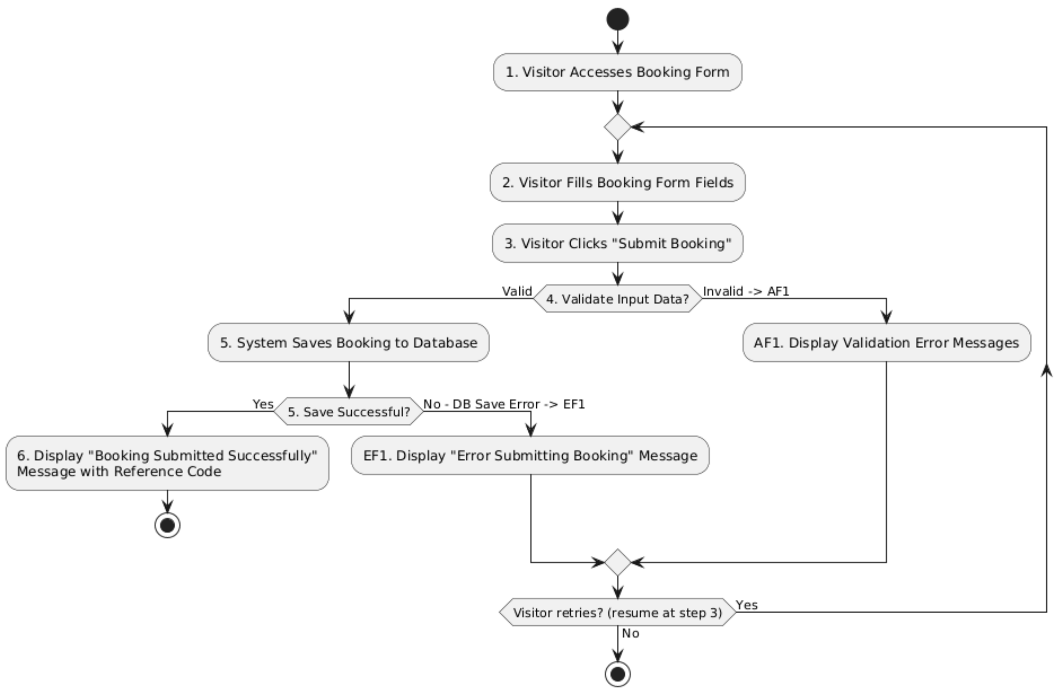


Figure 4.3: Activity Diagram for Submit Booking Inquiry (UC005)

4.3 Project Design

Al-Muneer Online Portal is an MVC Architecture-based platform operating through a single application only. The MVC architecture will ensure that there will be a well-defined division of roles including Model (including data and business logic), View (user interface), and Controller (managing user inputs and application flow). Such approach is extremely suitable for the requirements of the current project.

4.3.1 System Architecture Diagram

The below diagram illustrates the overall MVC architecture employed by the Al-Muneer Online Portal. In this combined system, the Spring Boot framework is used to handle the business logic as well as the User Interface design, creating an integrated and sustainable system.

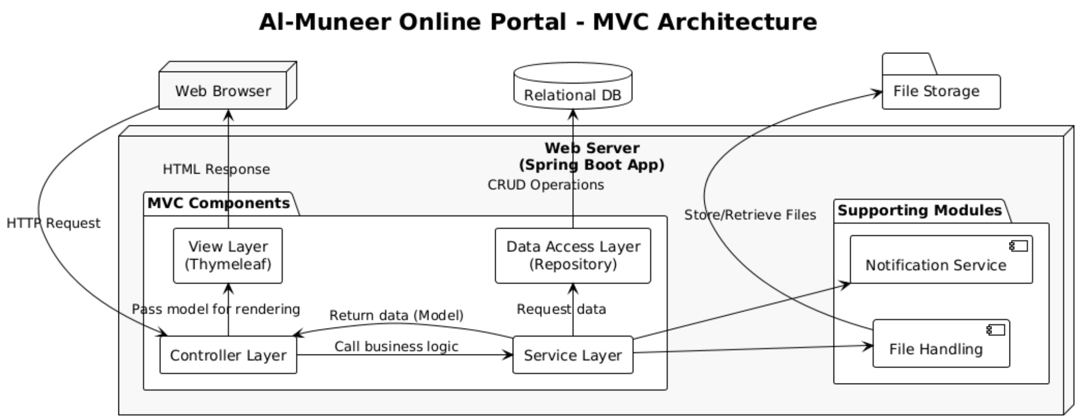


Figure 4.4: System Architecture Diagram (MVC)

4.3.2 Database Design

The reason for choosing the relational database model is due to the organized nature of the data. The following diagram is a basic ERD. The Primary Key (PK) and Foreign Key (FK) will be identified in the detailed design.

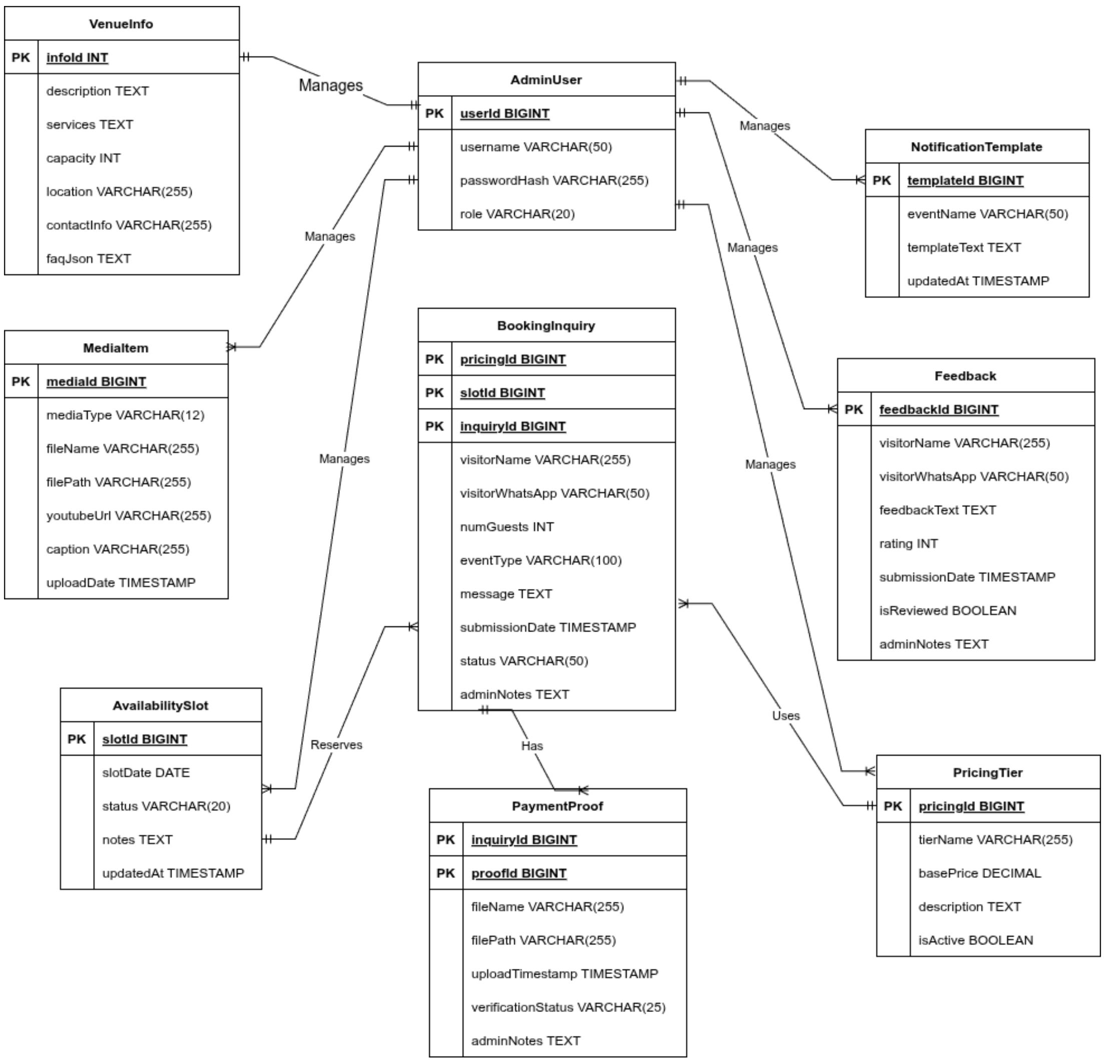


Figure 4.5: Conceptual Entity-Relationship Diagram (ERD)

Data Components:

- AdminUser: Stores admin auth and roles
- VenueInfo: General descriptive information about the venue
- MediaItem: Information regarding uploaded images and videos for the gallery
- AvailabilitySlot: Displays specific dates and whether they are booked or free.
- PricingTier : Details of various pricing packages or rates.
- BookingInquiry: Data about the inquiries of visitors.
- PaymentProof: Metadata of uploaded payment proof files, linked to inquiries.
- Feedback: Collects user submitted feedback.

4.4 Interface Design (Low-Fidelity)

4.4.1 Home Page

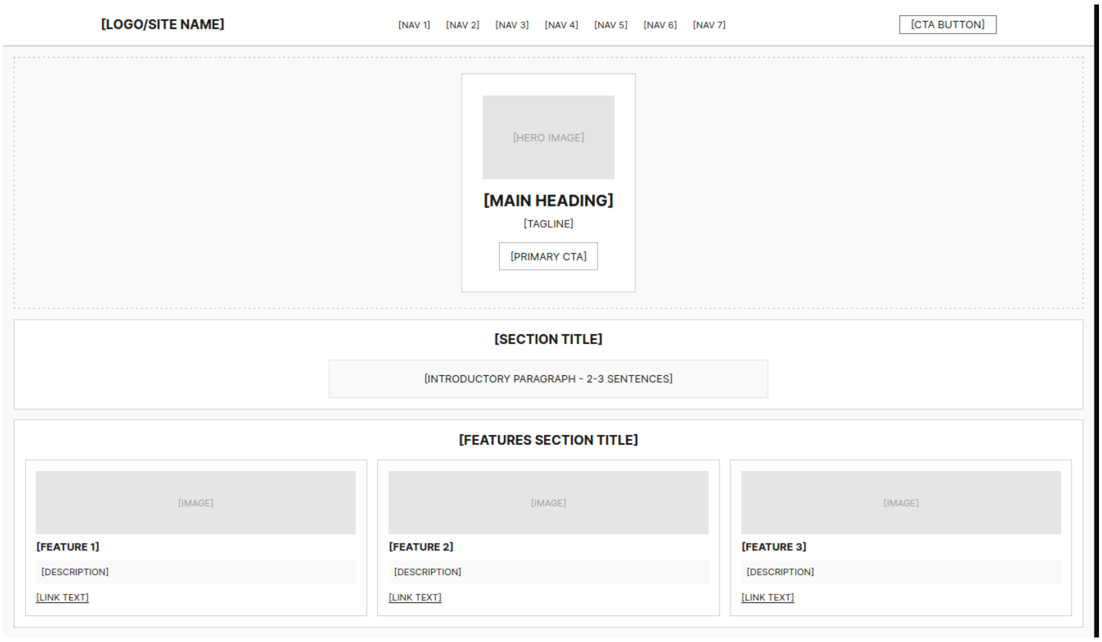


Figure 4.6: Low-Fidelity Wireframe: Homepage

4.4.2 Photo Gallery

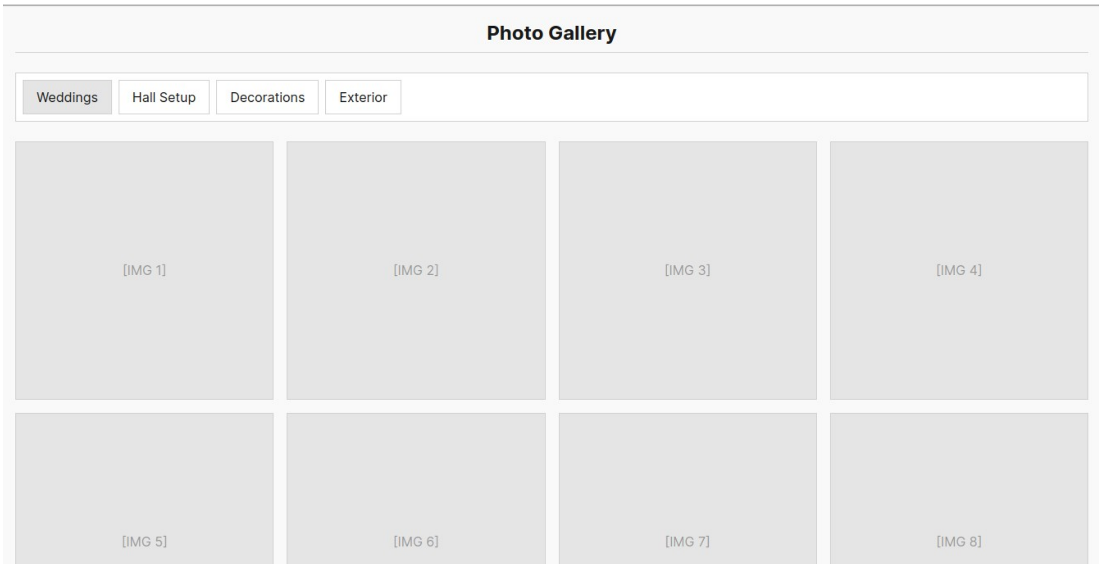


Figure 4.7: Low-Fidelity Wireframe: Gallery Page

4.4.3 Availability calendar

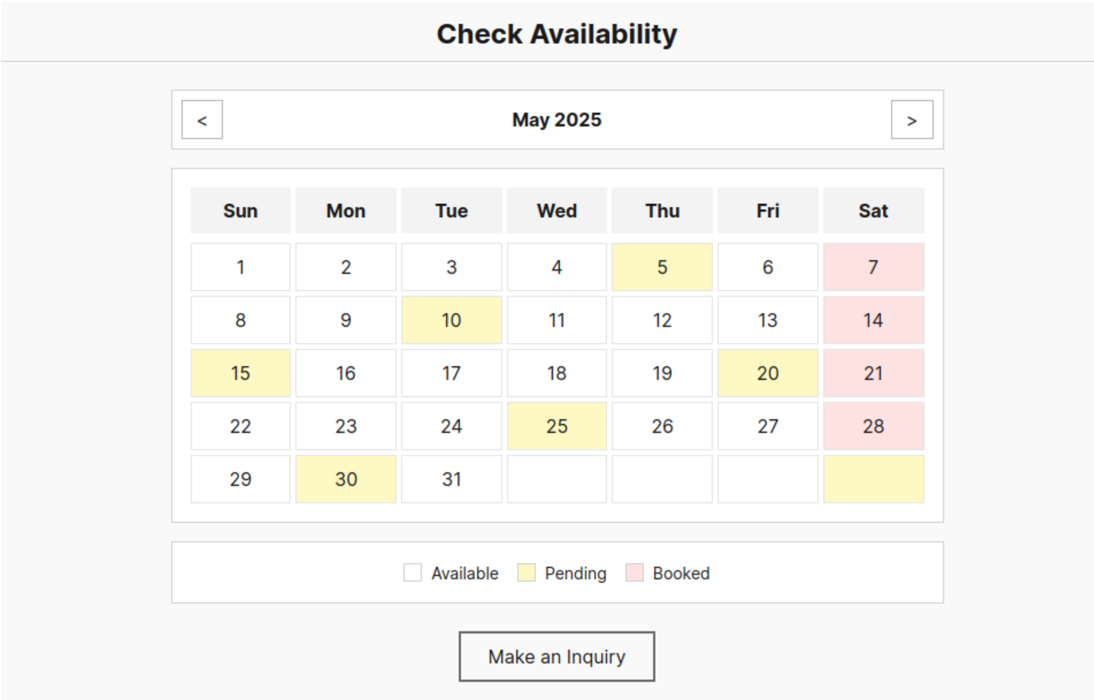


Figure 4.8: Low-Fidelity Wireframe: Availability Calendar Page

4.4.4 Pricing Page

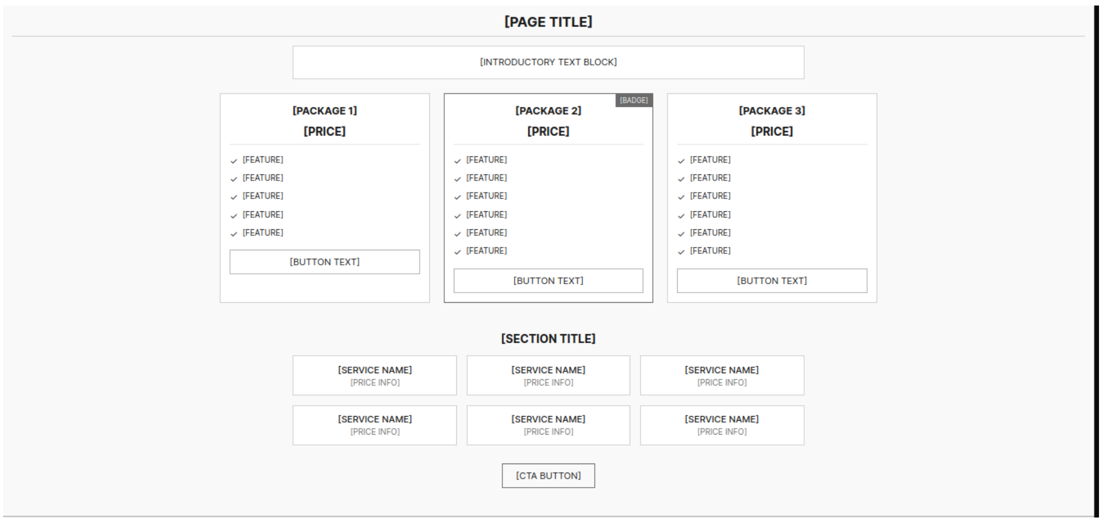


Figure 4.9: Low-Fidelity Wireframe: Pricing Page

4.4.5 Inquiry Form Page

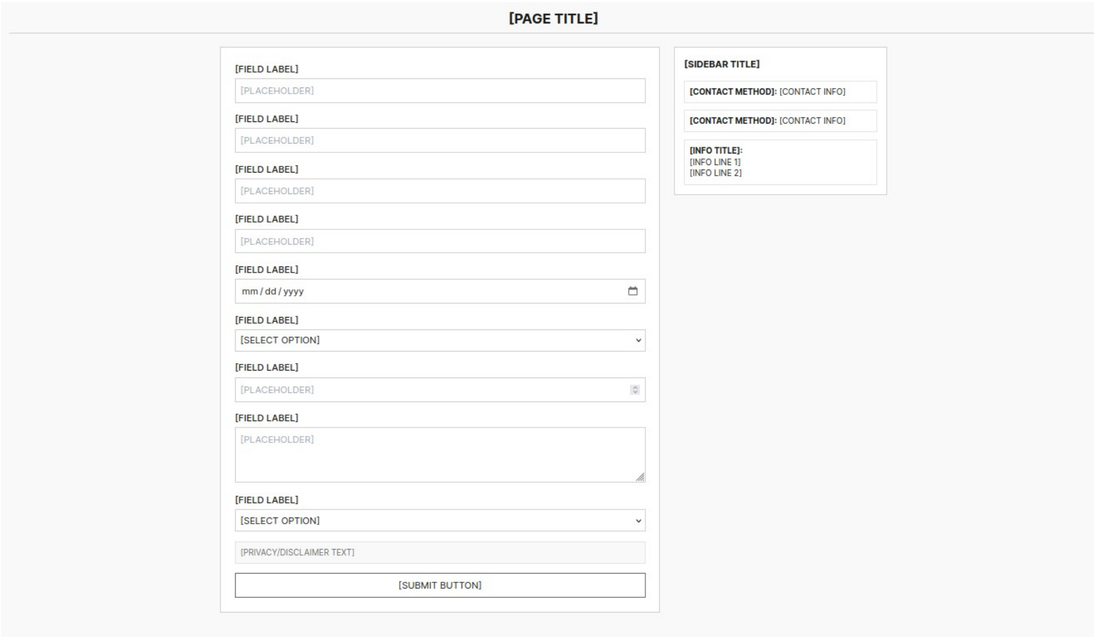


Figure 4.10: Low-Fidelity Wireframe: Inquiry Form Page

CHAPTER 5

RESULTS, TESTING AND DISCUSSION

5.1 Introduction

The following chapter highlights the implementation results of the Al-Muneer Online Portal after shifting from the PSM1 requirement design phase to the PSM2 implementation phase. The portal was developed using the Spring Boot framework and PostgreSQL database, authenticated with JWT and developed using Thymeleaf on the server side along with vanilla JS and Chart.js. The implementation process has been done using the Waterfall model which has been discussed in Chapter 3 while the requirements and design artifacts have been used from Chapter 3 and 4 respectively.

Discussion of the topic will be split into three major parts. The section 5.2 describes the process of coding of system major functionalities according to the changelog of the system implemented versions v0.3-v0.7. In section 5.3 there will be listed the key user interfaces, which show the results achieved by the system, including placeholders of screenshot descriptions. In section 5.4 there is described the process of testing performed, which includes black box testing, white box testing and user testing.

5.2 Coding of System's Main Functions

Backend for Al-Muneer Online Portal was built in Java 21 programming language using Spring Boot 3.4.4 framework, where the build tool is Maven. PostgreSQL 16 relational database system was used, and the security was provided using Spring Security framework with JWT token and BCrypt algorithm hashing. The frontend was created using HTML5, CSS3, Thymeleaf template engine, and plain JavaScript, with Chart.js library for visualisation of data. This section provides information about the main functional modules that have been programmed according to iterative changelog from v0.3 to v0.7.

5.2.1 Backend Foundation and Security

The backend architecture is a layered architecture: controllers, services, repositories and JPA entities . `AdminAuthController` is a controller to login user through Spring Security framework. The JWT tokens are stored in HTTP-only cookies for an authenticated session. Passwords are encrypted with BCrypt and the DataSeeder creates a default admin account during first start-up of the application. Domain classes are: `BookingInquiry`, `PaymentProof`, `AvailabilitySlot`, `PricingTier`, `Media`, `Feedback`, `PageVisit`, and `NotificationTemplate`.

AdminAuthController.java

```java
@Controller
@RequestMapping("/admin")
@RequiredArgsConstructor
public class AdminAuthController {

    private final AuthenticationManager authenticationManager;
    ⋮
    @PostMapping("/login")
    public String login(@RequestParam String username,
                        @RequestParam String password,
                        HttpServletResponse response,
                        Model model) {
        try {
            Authentication auth = authenticationManager.authenticate(
                    new UsernamePasswordAuthenticationToken(username, password));

            UserDetails userDetails = (UserDetails) auth.getPrincipal();
            String token = jwtUtil.generateToken(userDetails);

            Cookie cookie = new Cookie("jwt", token);
            cookie.setHttpOnly(true);
            cookie.setPath("/");
            cookie.setMaxAge(86400); // 24 hours
            response.addCookie(cookie);

            return "redirect:/admin/dashboard";
        } catch (AuthenticationException e) {
            model.addAttribute("error", "Invalid username or password");
            return "admin/login";
        }
    }
```

Figure 5.1: Code Snippet: Admin Login and JWT Cookie Issuance

5.2.2 Visitor Booking Flow and Reference Code System

The implementation’s main achievement is the uniform visitor booking process. The homepage has been converted into a one page scrolling landing page with anchor links. The hero section, venue details, media section, availability calendar and pricing packs have all been condensed onto a single page. Visitors interact through the interactive calendar where the future dates are with “Available” status and past dates are greyed out. The button “Submit Booking” when clicking on an available date submits the date to `/inquiry?date=YYYY-MM-DD`.

The booking landing page unifies two capabilities: submitting a new booking and retrieving existing bookings. Retrieval is designed so that visitors do not need to memorise any system-generated value: they simply enter the same WhatsApp number used at submission, and the system lists all of their bookings with current status. Each booking is additionally assigned a random 9-digit `referenceCode`, which serves as a secondary lookup key and is stored in a 150-day HTTP-only cookie named `inq_ref`, so returning visitors on the same device see their saved booking automatically. Visitors can also cancel their own booking directly from the confirmation page, provided no payment proof has been attached, which frees the associated availability slot.

```java
BookingInquiry.java

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long inquiryId;

    /**
     * Visitor-facing 9-digit reference code (e.g. 472918305).
     * Random, unique — used for lookup and cookie storage.
     * Nullable only to allow Hibernate schema-update on existing tables;
     * @PrePersist always populates this before insert.
     */
    @Column(unique = true)
    private Long referenceCode;
  ⋮
    @PrePersist
    protected void onCreate() {
        this.submissionDate = LocalDateTime.now();
        if (this.status == null) {
            this.status = InquiryStatus.NEW;
        }
        if (this.referenceCode == null) {
            // 9-digit random: 100_000_000 - 999_999_999
            this.referenceCode = ThreadLocalRandom.current().nextLong(100_000_000L, 1_000_000_000L);
        }
    }
```

Figure 5.2: Code Snippet: Booking Inquiry Submission Flow

5.2.3 Payment Proof, Status Cascade, and Notifications

The payment proof upload feature enables the visitor to submit their proof of offline payments in the form of images. The files are stored by the `FileUploadUtil` class in the `uploads/proofs/` directory while the file name in UUID format is saved in the `PaymentProof` class. Upon verifying the proof, an atomic cascade update is performed, making the payment proof Verified, and the `BookingInquiry` CONFIRMED along with `AvailabilitySlot` BOOKED.

WhatsApp notifications can be delivered dynamically using template-based systems. The system provides administrators the ability to add, edit and remove the notification templates within the admin section. Within the inquiry/payment detail view, the admin can use a dropdown list to select the notification template, preview the notification message with the placeholders replaced, and also generate the URL client-side for `wa.me`.

```java
PaymentProofServiceImpl.java

    @Override
    @Transactional
    public PaymentProof updateStatus(Long proofId, VerificationStatus status, String adminNotes) {
        PaymentProof proof = getById(proofId);
        proof.setVerificationStatus(status);
        if (adminNotes != null && !adminNotes.isBlank()) {
            proof.setAdminNotes(adminNotes);
        }
        PaymentProof saved = proofRepository.save(proof);

        // Cascade: if Verified → Inquiry=CONFIRMED, Slot=BOOKED
        if (status == VerificationStatus.VERIFIED) {
            BookingInquiry inquiry = saved.getInquiry();
            inquiry.setStatus(InquiryStatus.CONFIRMED);
            inquiryRepository.save(inquiry);

            if (inquiry.getSlot() != null) {
                inquiry.getSlot().setStatus(SlotStatus.BOOKED);
                slotRepository.save(inquiry.getSlot());
            }
        }

        return saved;
    }
```

Figure 5.3: Code Snippet: Payment Verification Status Cascade

5.2.4 Administrator Modules and Analytics

The admin dashboard offers a daily workload snapshot in terms of new and active inquiries, pending payment proofs, unreviewed feedback, today's and past seven days' visits, average feedback rating, recent inquiries, and popular pages. Analytics shows the website traffic statistics by means of Chart.js. There is a bar chart for popular pages and line chart for daily trends of the past 30 days on the analytics page. Reports offer pie and bar charts to show the statuses of the inquiries, payments, and feedback ratings within the chosen time period by means of `?fromDate=` and `?toDate=` query parameters.

```java
    public DashboardStats getStats() {
        long newInquiries = inquiryRepository.countByStatus(InquiryStatus.NEW);
        long activeInquiries = inquiryRepository.countByStatusNotIn(
                List.of(InquiryStatus.COMPLETED, InquiryStatus.CANCELLED_BY_ADMIN, InquiryStatus.CANCELLED_BY_VISITOR));

        long pendingPayments = paymentProofRepository.countByVerificationStatus(
                VerificationStatus.PENDING_VERIFICATION);

        long unreviewedFeedback = feedbackRepository.countByIsReviewed(false);

        LocalDate today = LocalDate.now();
        long visitsToday = sumHitsForDay(today);
        long visitsLast7Days = sumHitsSince(today.minusDays(6));

        double avgRating = feedbackRepository.findAverageRating();

        List<BookingInquiry> recent = inquiryRepository.findAllByOrderBySubmissionDateDesc()
                .stream().limit(5).toList();

        Map<String, Long> topPages = new LinkedHashMap<>();
        pageVisitRepository.findTotalHitsPerPage().stream().limit(5).forEach(row ->
                topPages.put((String) row[0], ((Number) row[1]).longValue()));

        return new DashboardStats(
                newInquiries,
                activeInquiries,
                pendingPayments,
                unreviewedFeedback,
                visitsLast7Days,
                visitsToday,
                avgRating,
                recent,
                topPages
        );
    }
```

Figure 5.4: Code Snippet: Admin Dashboard Statistics Aggregation

5.2.5 AI Integration with Gemini

Version v0.7 added three non-blocking AI advisors powered by the Gemini model and the `google-genai` Java SDK (version 1.56.0):

- **AI Business Report Advisor** derives booking conversion and cancellation rates and prompts Gemini for three number backed action bullets.
- **AI Feedback Advisor** is separating low rating complaints from positive ones so the owner should address first.
- **AI Traffic Funnel Advisor** maps Home → Gallery → Pricing → Inquiry visit counts to see funnel drop off and suggest concrete conversion improvement.

All the AI panels are asynchronously loaded via dedicated `GET /ai-insight` endpoints after the page is rendered, so the main interface remains fast and can degrade gracefully if the API is unavailable.

5.3 Essential Interfaces Showing System's Results and Achievements

This section describes the main interfaces that demonstrate the functional aspects of the developed solution. Screenshots from the actual system are used to fill the placeholders in the final report.

5.3.1 Visitor Home Page and Booking Calendar

Visitor home page has one page scroll design. First up is the hero section, then the venue info which includes a Google Maps iframe, a masonry gallery with filters based on labels, and an availability calendar and pricing packages. Also there is anchor navigation with scroll spy to highlight the current section. The calendar allows the user to change months, shows the difference between available and booked days and activates a context menu “Submit Booking” when an available day is clicked by the visitor.

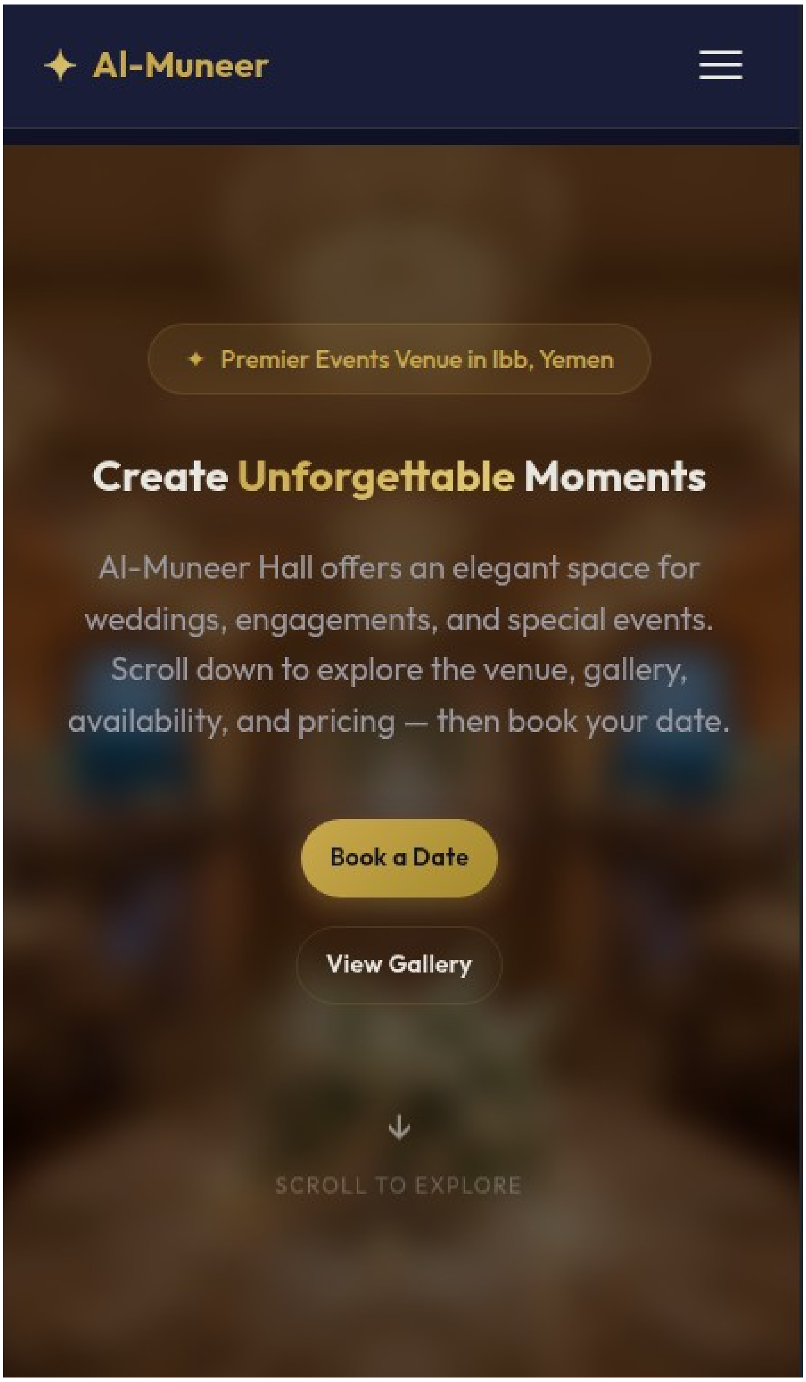


Figure 5.5: Visitor Home Page with Single-Page Scroll Layout

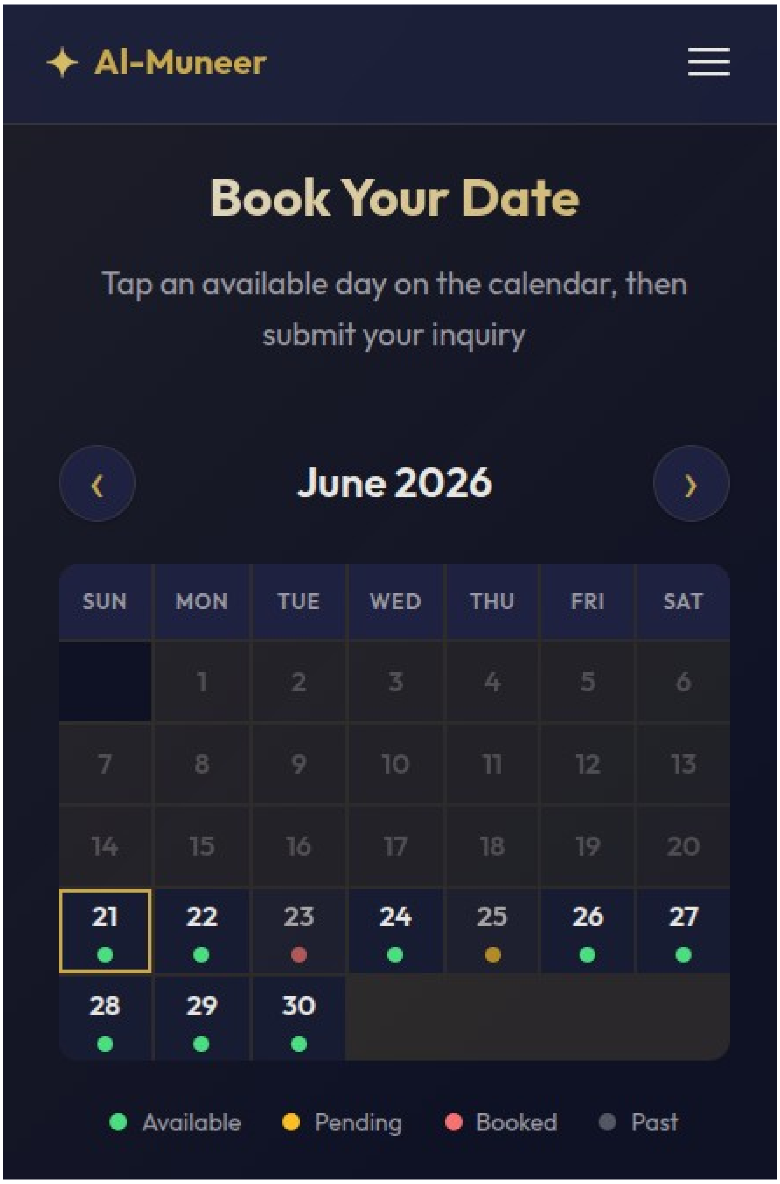


Figure 5.6: Visitor Availability Calendar and Inquiry CTA

5.3.2 Inquiry Form and Reference Code Lookup

The /inquiry page serves both new enquiries as well as lookup enquiries. When accessed from the homepage, the user is given prefilled fields in the form for the date and pricing package. The visitor receives a confirmation page with the 9-digit reference code and the option to cancel the inquiry themselves after submitting the form. The same page has a lookup field for the reference code.

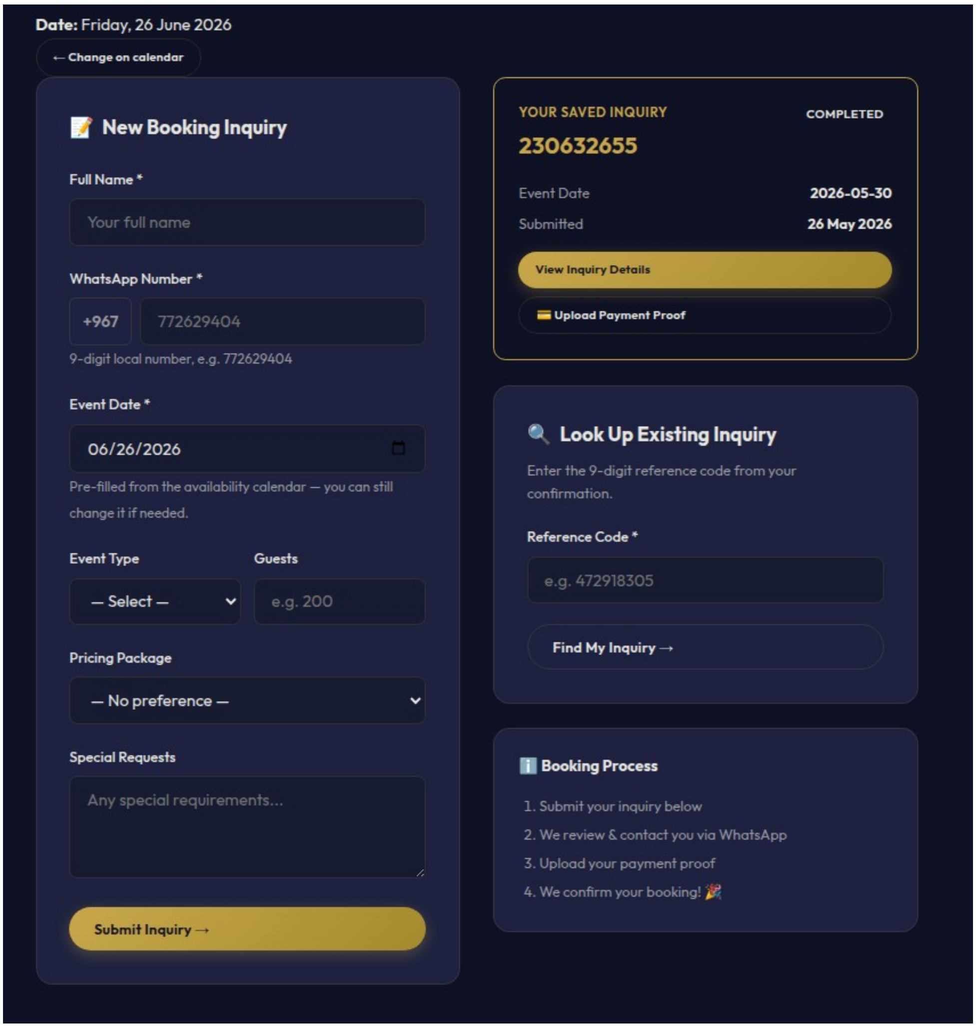


Figure 5.7: Inquiry Form with Pre-filled Date and Package

5.3.3 Admin Dashboard and Analytics

The admin dashboard provides an overview of the day-to-day business of the company to the owner. These are pending/active requests, proof pending, feedback not reviewed, and traffic history. The analytics page is further complemented by visualisation through the Chart.js graphs and AI Traffic Funnel Advisor.

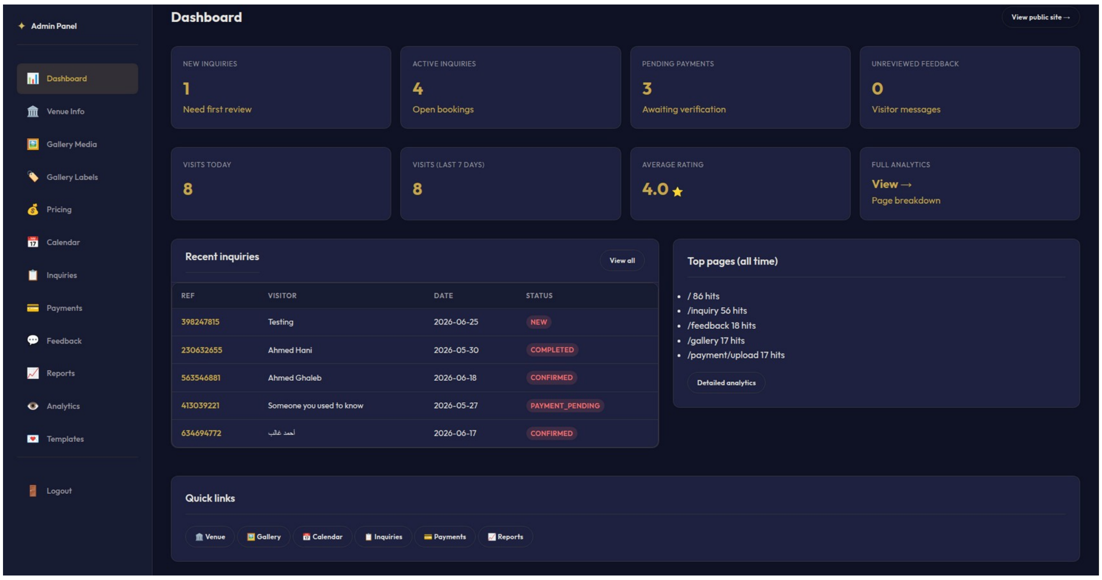


Figure 5.8: Administrator Dashboard Overview

5.3.4 Admin Inquiry and Payment Management

The status filter chip along with per-status count, client-side search and “only active” default view is part of the inquiry management panel. The information shown in each row includes the visitor’s reference code, contact details, date of the event, and the status of the request. In the detail view, the administrator has options for updating status, viewing the attached payment proofs relevant to that inquiry, and sending a dynamic WhatsApp message from a list of templates.

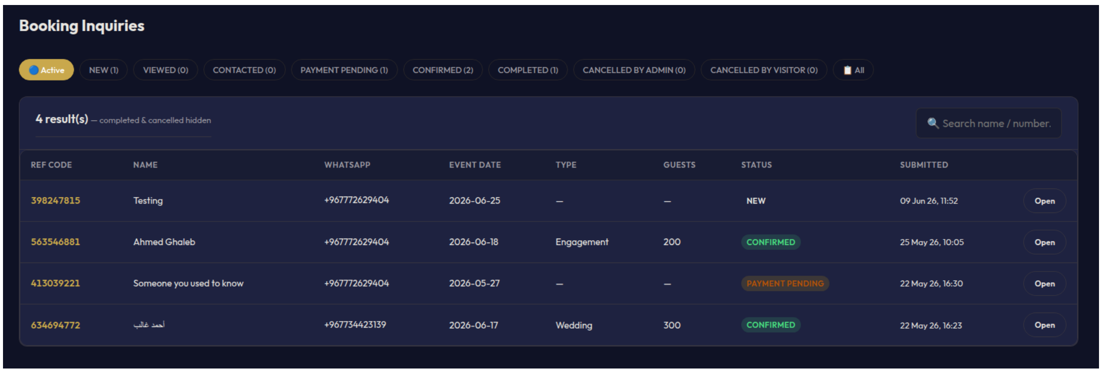


Figure 5.9: Administrator Inquiry Management Panel

5.3.5 Reports with Visualisations

The Reports section depicts operational information using visual diagrams. The pie charts reveal information about inquiry status and payment status. A bar chart reveals information on feedback ratings. Users can use the date filter to view the trends during a certain period. The AI Business Report Advisor section gets loaded asynchronously for useful suggestions on the basis of current report information.

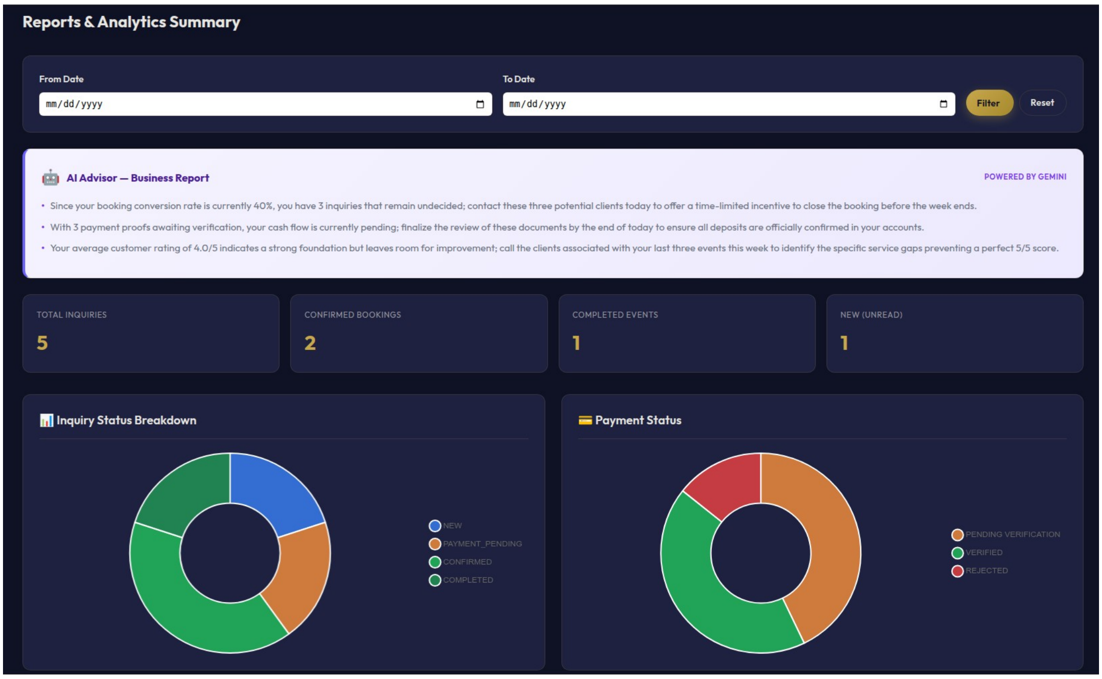


Figure 5.10: Reports Page with Visual Charts

Figure 5.11: Analytics Dashboard with Chart.js Visualizations

5.4 Testing

Testing is carried out during the implementation stage to ensure that the Al-Muneer Online Portal satisfies the functional and non-functional requirements stated in the SRS. The methods used included black box, white box, and user testing.

5.4.1 Black Box Testing

The black-box testing was concerned with flows within the system, input/output behavior and errors without looking into the source code. The test cases have been generated using the use cases in the SRS document and changelog features:

- **Visitor flow:** Submission of valid/invalid inquiry form, pre-fill date/package information from homepage, search for inquiries through reference number, and cancellation of inquiries.
- **Payment proof flow:** Uploading valid and invalid file types, verifying a proof as an admin, and confirming the status cascade to inquiry and slot.
- **Admin flow:** Login, management of venues/gallery label names, updates to calendar slots, inquiry filtration, messaging through WhatsApp templates, and report generation.
- Error handling: Invalid login credentials, missing required fields, oversized uploads, and unauthorized access attempts all produced clear user-friendly messages.

The black box tests confirmed that the system behaves according to specification and handles edge cases gracefully.

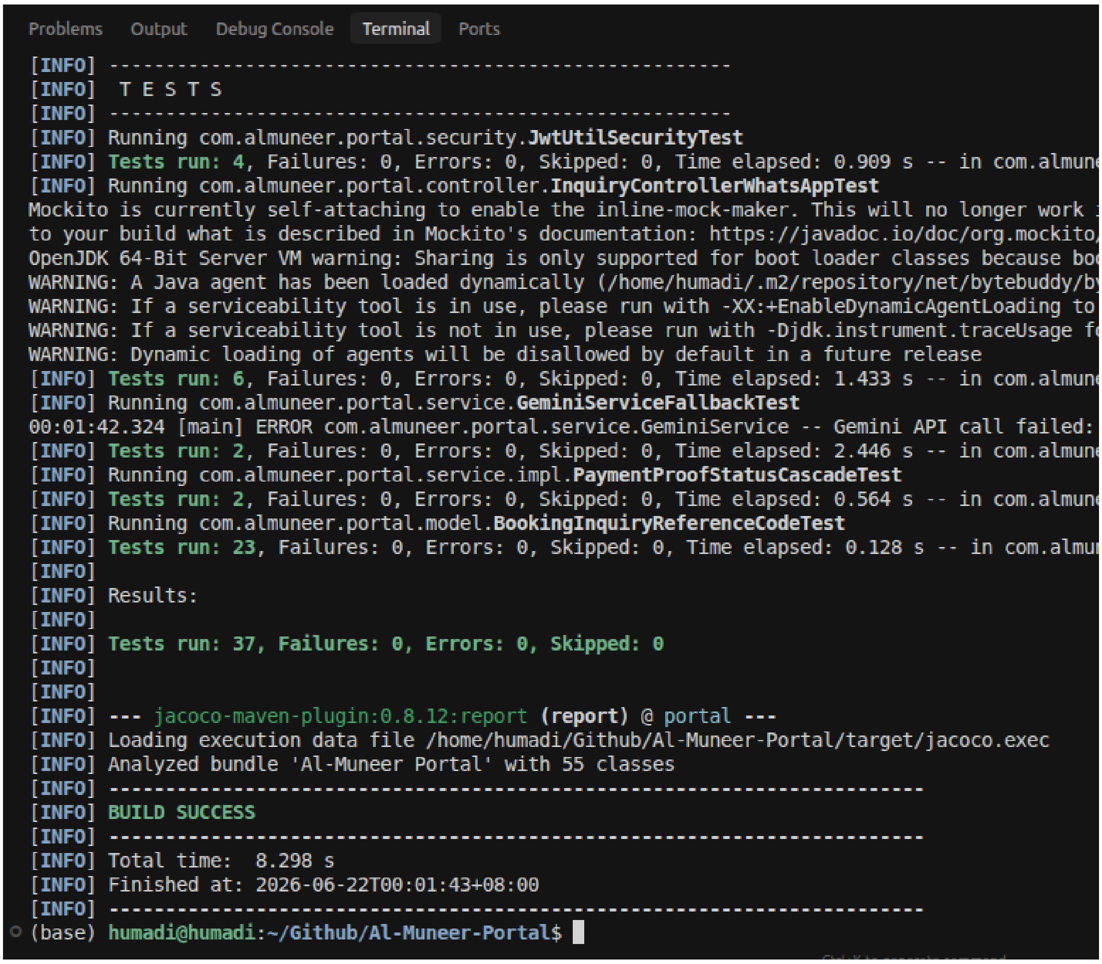
5.4.2 White Box Testing

White box testing was used to test the internal logic and code paths. The following components were unit and integrated tested:

- Reference code generation: We confirmed that a unique 9-digit code is assigned to `BookingInquiry`.
- WhatsApp normalisation: `normaliseWhatsApp()` is known to remove non-digits and prefix the configured country code.
- Status cascade: Confirmed that when a `PaymentProof` is checked, the corresponding `BookingInquiry` is set to CONFIRMED and the `AvailabilitySlot` is set to BOOKED.
- JWT security: Token creation, validation and expiration management.

AI service fallback: Verified `GeminiService` provides a graceful fallback message on API failure without crashing the page.

confirmed that the core business rules and security mechanisms are working correctly at the code level.



```
Problems   Output   Debug Console   Terminal   Ports
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.almuneer.portal.security.JwtUtilSecurityTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.909 s -- in com.almune
[INFO] Running com.almuneer.portal.controller.InquiryControllerWhatsAppTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work
to your build what is described in Mockito's documentation: https://javadoc.io/doc/org.mockito/
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because boo
WARNING: A Java agent has been loaded dynamically (/home/humadi/.m2/repository/net/bytebuddy/by
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage fo
WARNING: Dynamic loading of agents will be disallowed by default in a future release
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.433 s -- in com.almune
[INFO] Running com.almuneer.portal.service.GeminiServiceFallbackTest
00:01:42.324 [main] ERROR com.almuneer.portal.service.GeminiService -- Gemini API call failed:
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.446 s -- in com.almune
[INFO] Running com.almuneer.portal.service.impl.PaymentProofStatusCascadeTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.564 s -- in com.almune
[INFO] Running com.almuneer.portal.model.BookingInquiryReferenceCodeTest
[INFO] Tests run: 23, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.128 s -- in com.almun
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 37, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco-maven-plugin:0.8.12:report (report) @ portal ---
[INFO] Loading execution data file /home/humadi/Github/Al-Muneer-Portal/target/jacoco.exec
[INFO] Analyzed bundle 'Al-Muneer Portal' with 55 classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.298 s
[INFO] Finished at: 2026-06-22T00:01:43+08:00
[INFO] ------------------------------------------------------------------------
(base) humadi@humadi:~/Github/Al-Muneer-Portal$
```

Figure 5.9: Unit Testing Log

5.4.3 User Testing

Project stakeholder Mr. Ahmed Almunajid owner of Al-Muneer Hall participated in user testing. The owner performed representative duties for the website such as updating venue information, uploading gallery pictures, looking at new enquiries, looking at a payment proof, and looking at reports. Interface clarity, workflow efficiency and usefulness of the AI advisors were collected as feedback. Minor changes to dashboard card ordering and notification template defaults were made based on this feedback.

5.5 Chapter Summary

This chapter presents the results of the implementation of the Al-Muneer Online Portal. Section 5.2 highlighted the evolution of core functions like the Spring Boot back end, visitor reservation process using codes, cascade of payment evidence verification, automated notifications via WhatsApp, dashboards for administrators, analytical data, reports, and Gemini AI advisors. Section 5.3 described the critical interfaces and Section 5.4 discussed the black-box, white-box and user testing. The system developed clearly satisfies all requirements in the SRS and provides a practical approach.

CHAPTER 6
CONCLUSION

6.1 Introduction

This marks the end of the report on the development of the Al-Muneer Online Portal project. The project went from the PSM1 stage to the PSM2 stage where a working online portal for Al-Muneer Hall was developed. The implementation of the system, together with the interfaces used and the tests carried out, were all described in Chapter 5. This chapter highlights the accomplishments of the project as well as possible future directions for improvement and maintenance.

6.2 Achievement of Project Objectives

The four project objectives defined in Section 1.4 have been addressed throughout the project lifecycle. The first objective, requirements elicitation and analysis, was completed in PSM1 through stakeholder engagement with Mr. Ahmed Almunajid, producing a comprehensive Software Requirements Specification (SRS). The second objective, system design, was fulfilled and documented in the Software Design Description (SDD), including the MVC architecture, database schema, and UI/UX wireframes.

During PSM2, the third objective, system development, was realised by building the portal with Spring Boot, PostgreSQL, Thymeleaf, and JavaScript. All core modules were implemented: visitor information browsing, media gallery with label filters, interactive availability calendar, dynamic pricing display, booking submission with 9-digit reference codes, payment proof upload, feedback submission, JWT-secured administrator authentication, content management, booking and payment management, WhatsApp notification templates, calendar management, dashboard analytics, reports with charts, and Gemini AI advisors.

The fourth objective, testing and validation, was achieved through black box, white box, and user testing, as detailed in Chapter 5 and the Software Test Documentation

(STD). The system was found to operate reliably and to meet the functional and non-functional requirements. In addition to the defined objectives, the portal was deployed to a Cloud VPS, making it accessible via standard web browsers, and technical documentation, user guidance, and maintenance considerations were prepared to support future upkeep.

6.3 Future Improvements

While the current design meets the desired scope, the following improvements may be considered for later developments:

- **Direct Online Payments:** Integrating a formal payment gateway would remove reliance on manual transfer screenshots and provide immediate booking confirmation.
- **WhatsApp Business API:** The use of WhatsApp Business API instead of client side deep links would provide more options for automated and two way messaging notifications.
- **Multi-Language Support:** Adding full Arabic and English localisation would improve accessibility for a broader audience.
- **Mobile Application:** A companion mobile app could offer push notifications and a native experience for frequent users.

This will be built on the already established strong base of the Al Muneer Online Portal.

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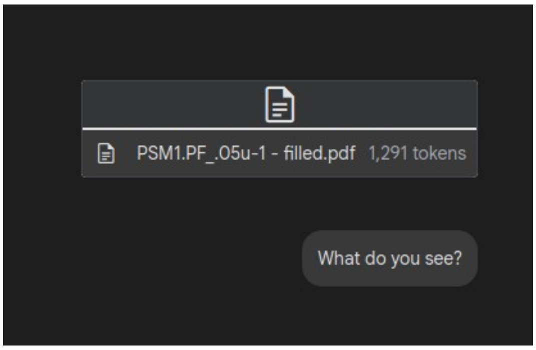
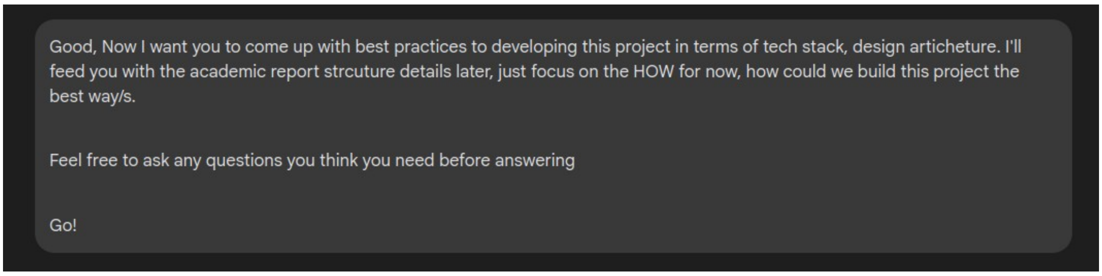
Appendix A - List of Prompts

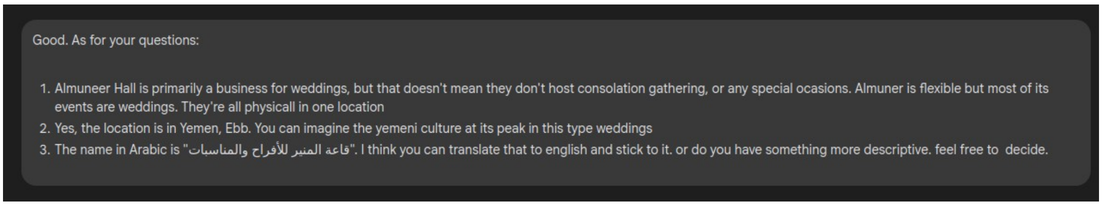
1. Giving AI context before brainstorming different solutions



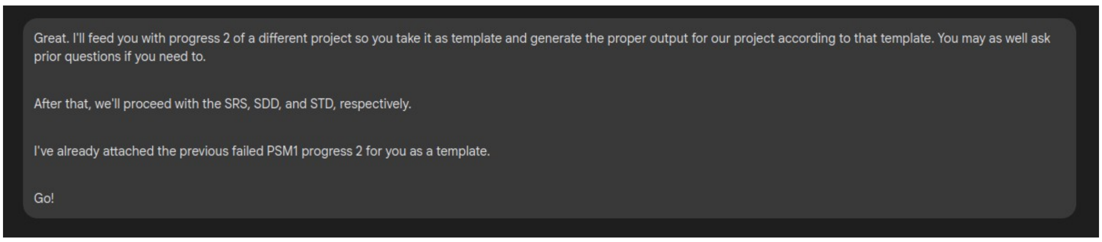
2. Telling AI to ask question to get more context if needed



3. Answering AI questions



4. Getting properly structured output



5. Text to Gantt Chart design. Let the AI do the work!